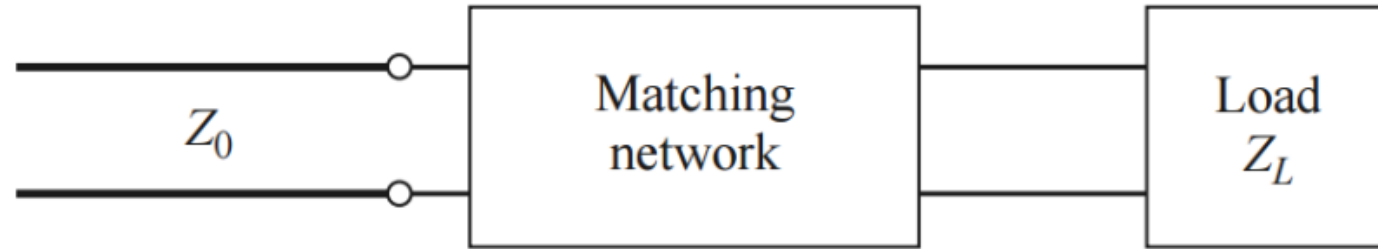


Single Stub Matching

Impedance Matching

- What?



- Why?

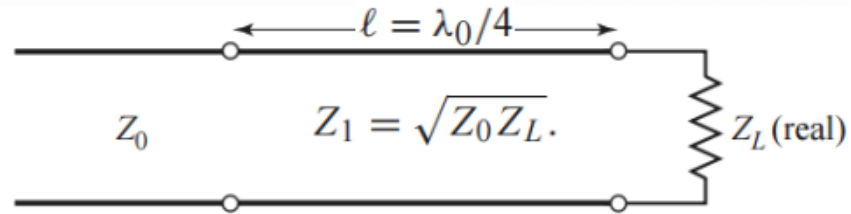
- Maximum power delivery
- Improves SNR (antennas, low-noise amplifier, etc.)

- Key factors:

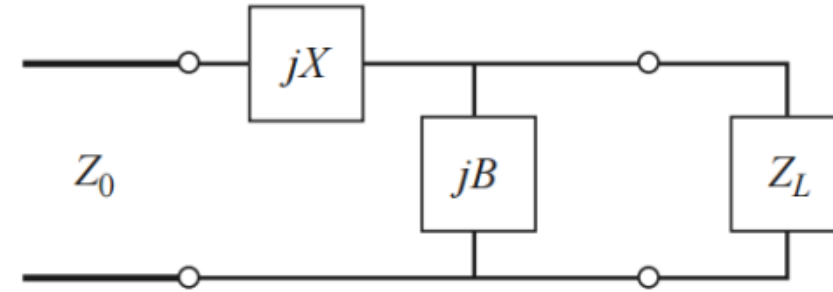
- Complexity (simple $\Rightarrow$ cheaper $\Rightarrow$ less lossy)
- Bandwidth (MN can ideally give $\Gamma=0$ at single frequency)
- Implementation (Waveguide, coaxial, microstrip line, etc.)
- Adjustability (sometimes we have variable load impedance)

Impedance Matching

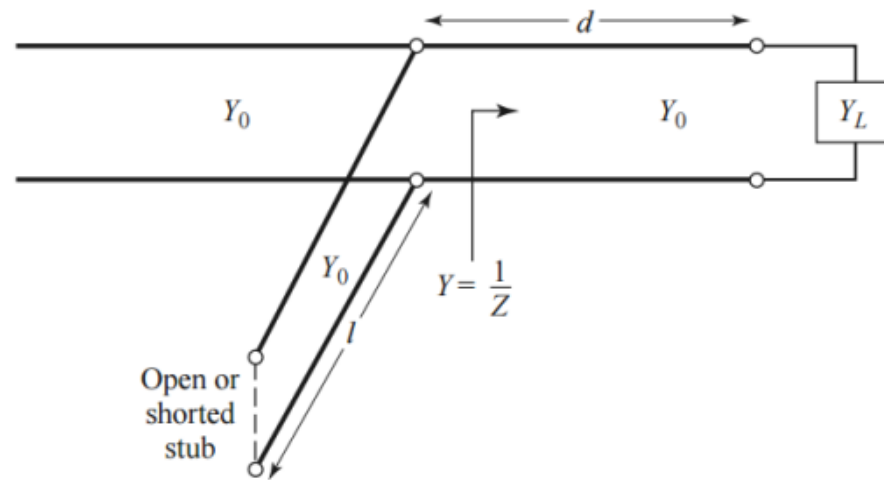
- Techniques:



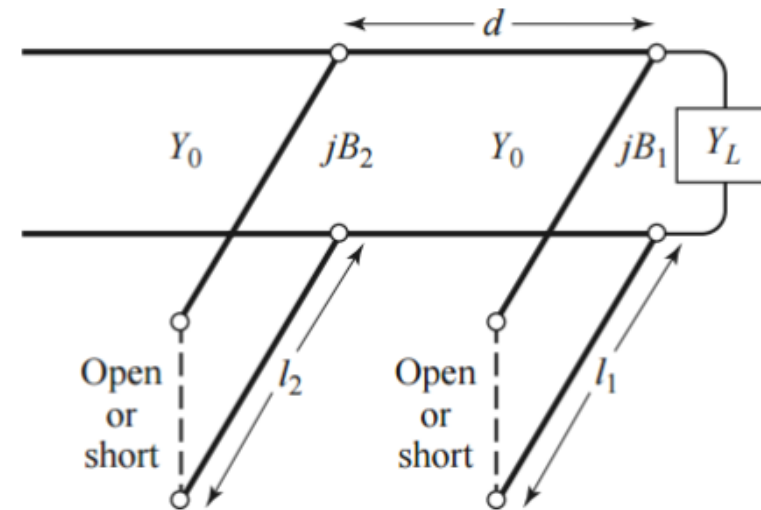
Quarter-Wave Matching (For real Z_L)



Matching with Lumped components



Single-Stub Matching



Double-Stub Matching

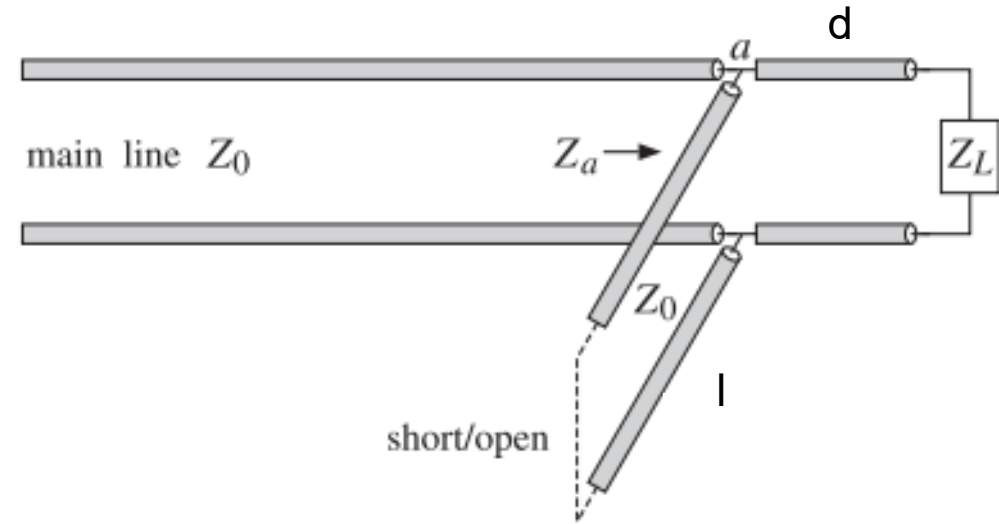
Single Stub Matching

Stub tuners are widely used to match any complex load to a main line. They consist of shorted or opened segments of the line, connected in parallel or in series with the line at a appropriate distances from the load. For this course we will focus on the parallel (shunt).

In coaxial cable or two-wire line applications, the stubs are obtained by cutting appropriate lengths of the main line.

Short Vs Open Stubs (tuners): Shorted stubs are usually preferred because opened stubs may radiate from their opened ends.

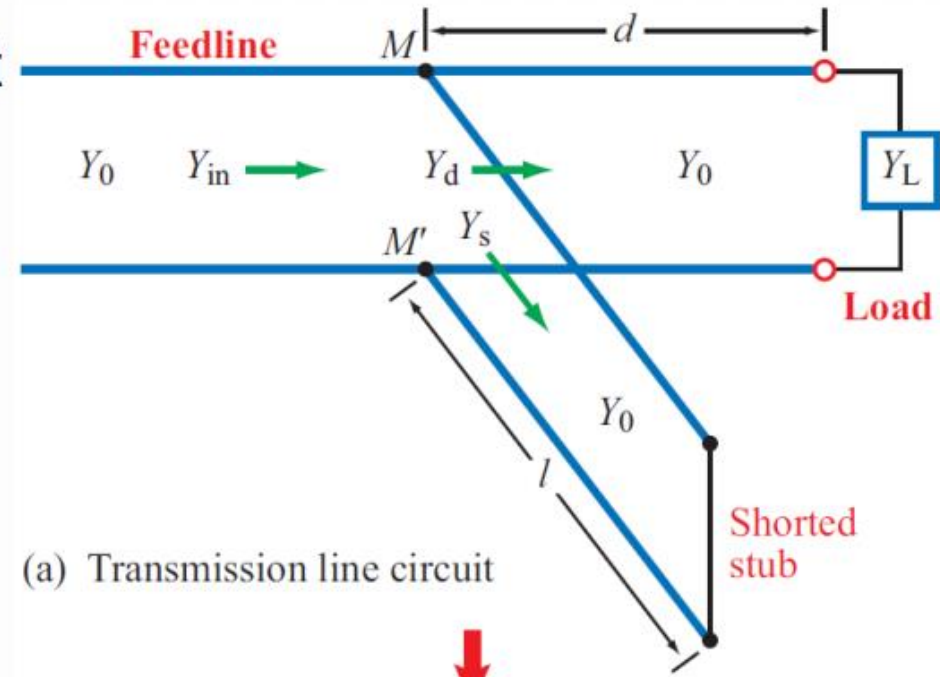
Limitation: The resistive part of the load must be non-zero. Purely reactive loads cannot be matched to a real line impedance by this method.



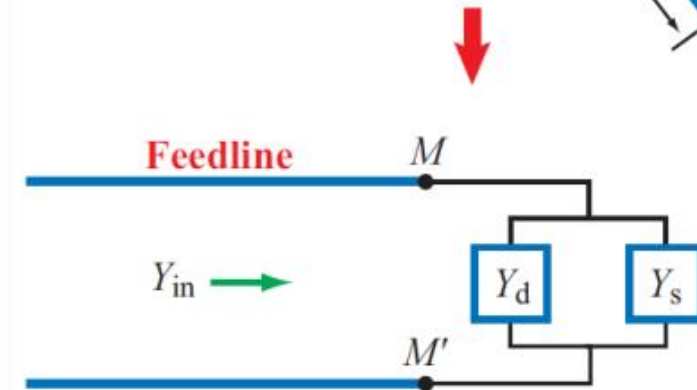
Single-Stub Matching

- For the shunt-stub case:
 - The basic idea is to select d so that the admittance, Y_d , seen looking into the line at distance d from the load is of the form $Y_0 + jB$ (or $1 + jb$ after normalization)
 - Then, the stub susceptance is chosen as $-jB$. Resulting in a matched condition.

Goal : $Y_{in} = Y_d + Y_s = Y_0$

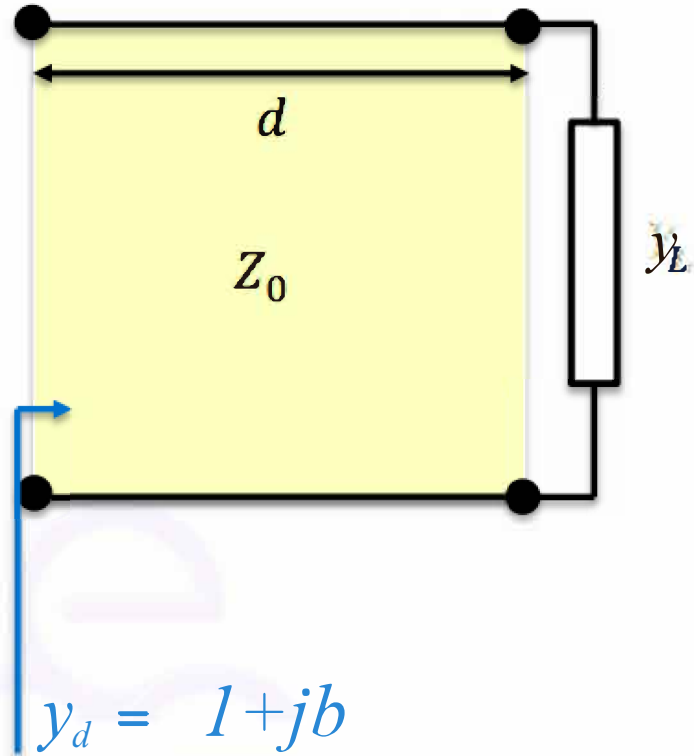


(a) Transmission line circuit



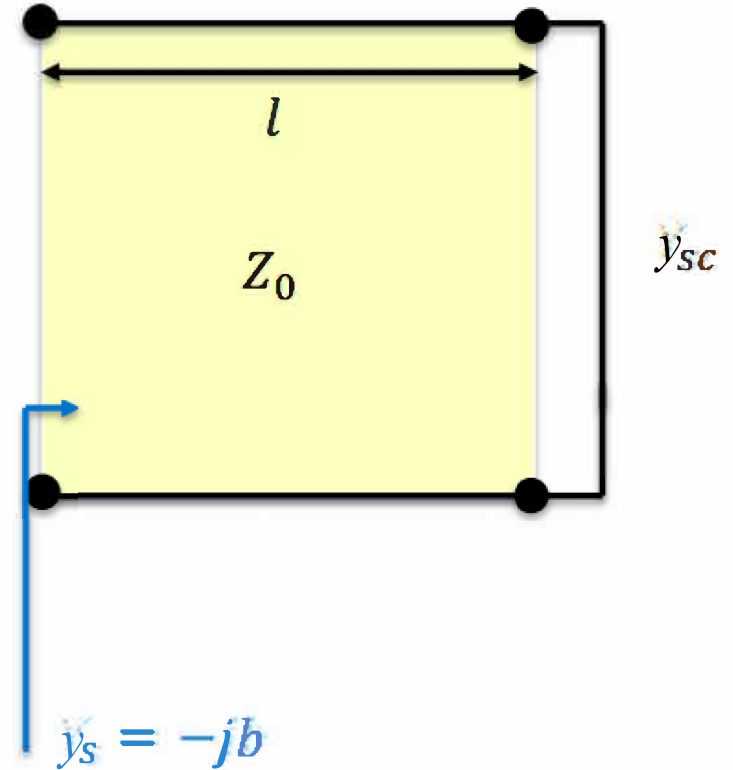
(b) Equivalent circuit

Two Design Milestones



1

Find d such that input admittance is on the form $1 + jb$ & find b



2

Find l such that input admittance is on the form $-jb$

Single-Stub Matching

- Smith Chart Steps:

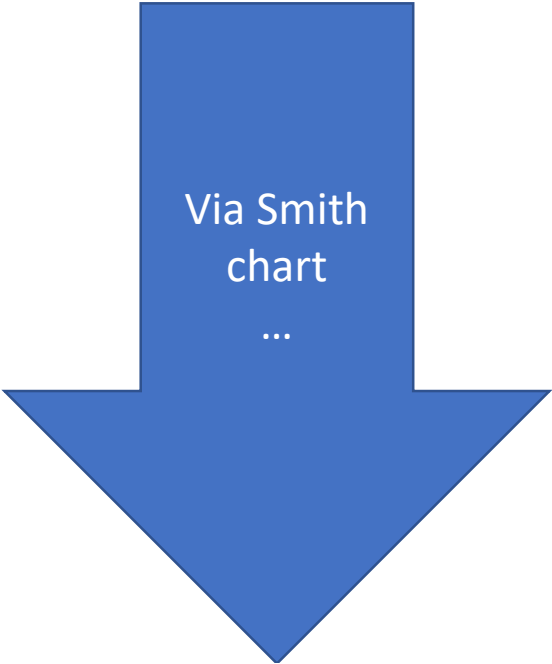
1. Plot the normalized load impedance $z_L = Z_L/Z_0$
2. Construct the appropriate SWR circle and convert to the load admittance, y_L
3. For the remaining steps we consider the Smith chart as an admittance chart
4. Notice that the SWR circle will intersect the $1 + jb$ circle at two points.
Therefore the distance d from the load to the stub is given by either of the two intersections.
5. Read the WTG (Wavelengths towards generator) scale to get two possible solutions of d
6. Get the normalized admittances at the two intersection points (should be in the form of $1 + jb$)
7. Therefore, we require a stub with a susceptance of $-jb$, the length of a shorted stub (or opened) that gives this susceptance can be found on the Smith chart by starting at $y = \infty$ (the short circuit) and moving along the outer edge of the chart toward the generator to the $-jb$ point.
8. Reading the scale, you can get l .

Example:

A load impedance $Z_L = 25 - j50 \, \Omega$ is connected to a $50 \, \Omega$ transmission line. Insert a shunt element to eliminate reflections towards the sending end of the line. Specify the insert location d (in wavelengths), the type of element, and its value, given that $f = 100 \text{ MHz}$.

The normalized load impedance is

$$z_L = \frac{Z_L}{Z_0} = \frac{25 - j50}{50} = 0.5 - j1,$$



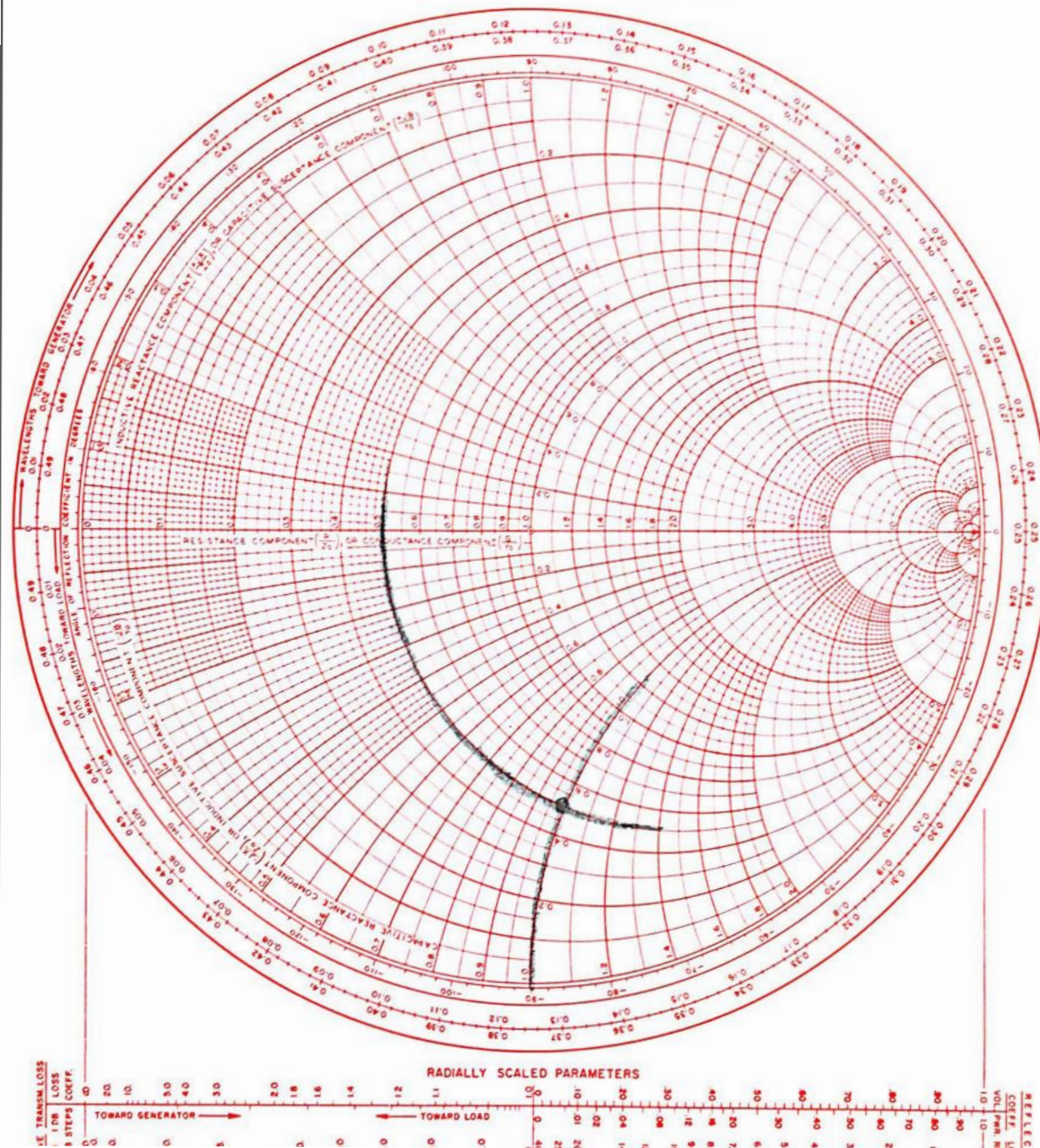
Via Smith
chart
...

The normalized load admittance is

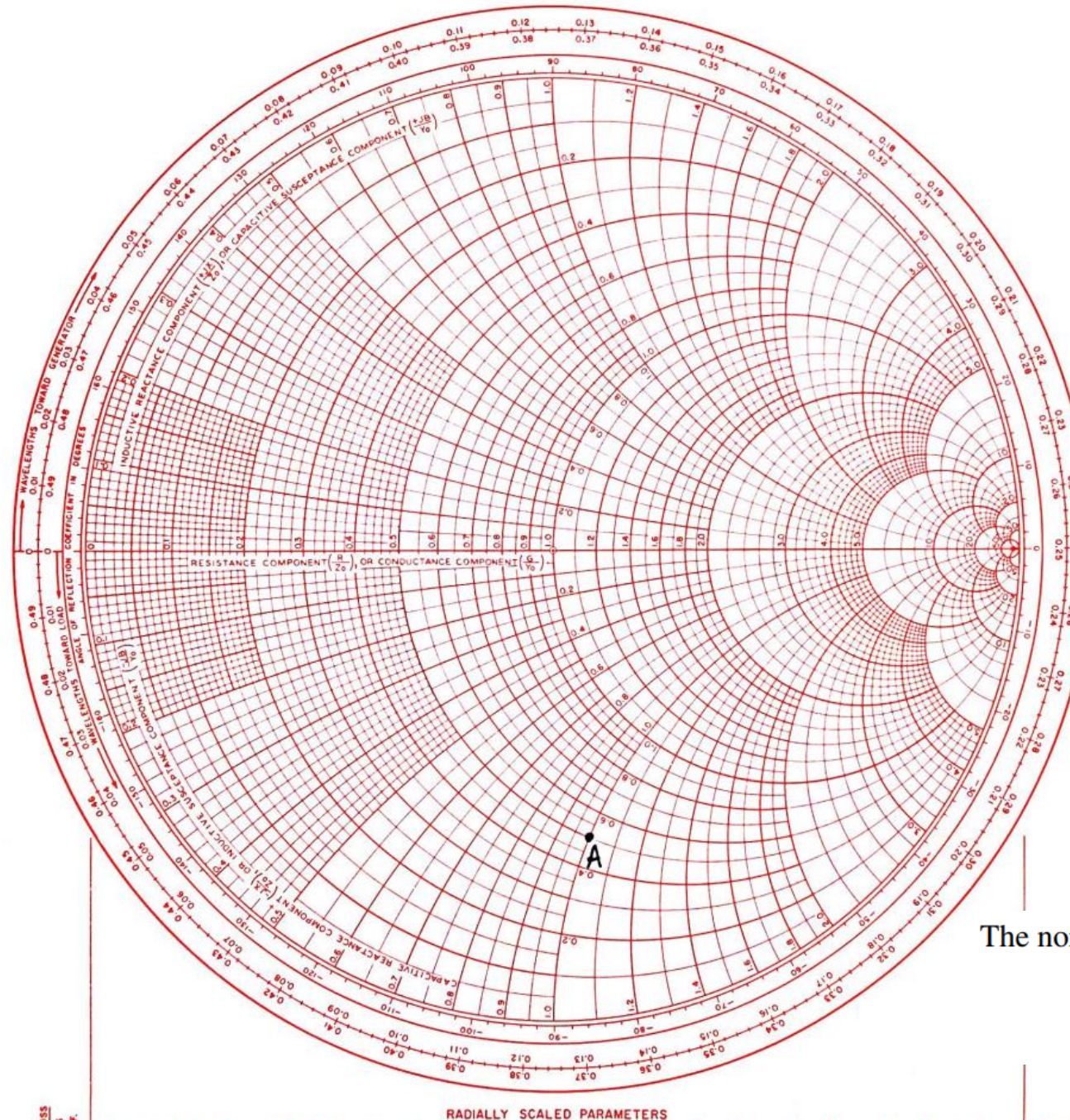
$$y_L = \frac{Z_0}{Z_L} = 50 \left(\frac{1}{25 - j50} \right) = 0.4 + j0.8.$$

$$y_L = 0.4 + j0.8.$$

IMPEDANCE OR ADMITTANCE COORDINATES



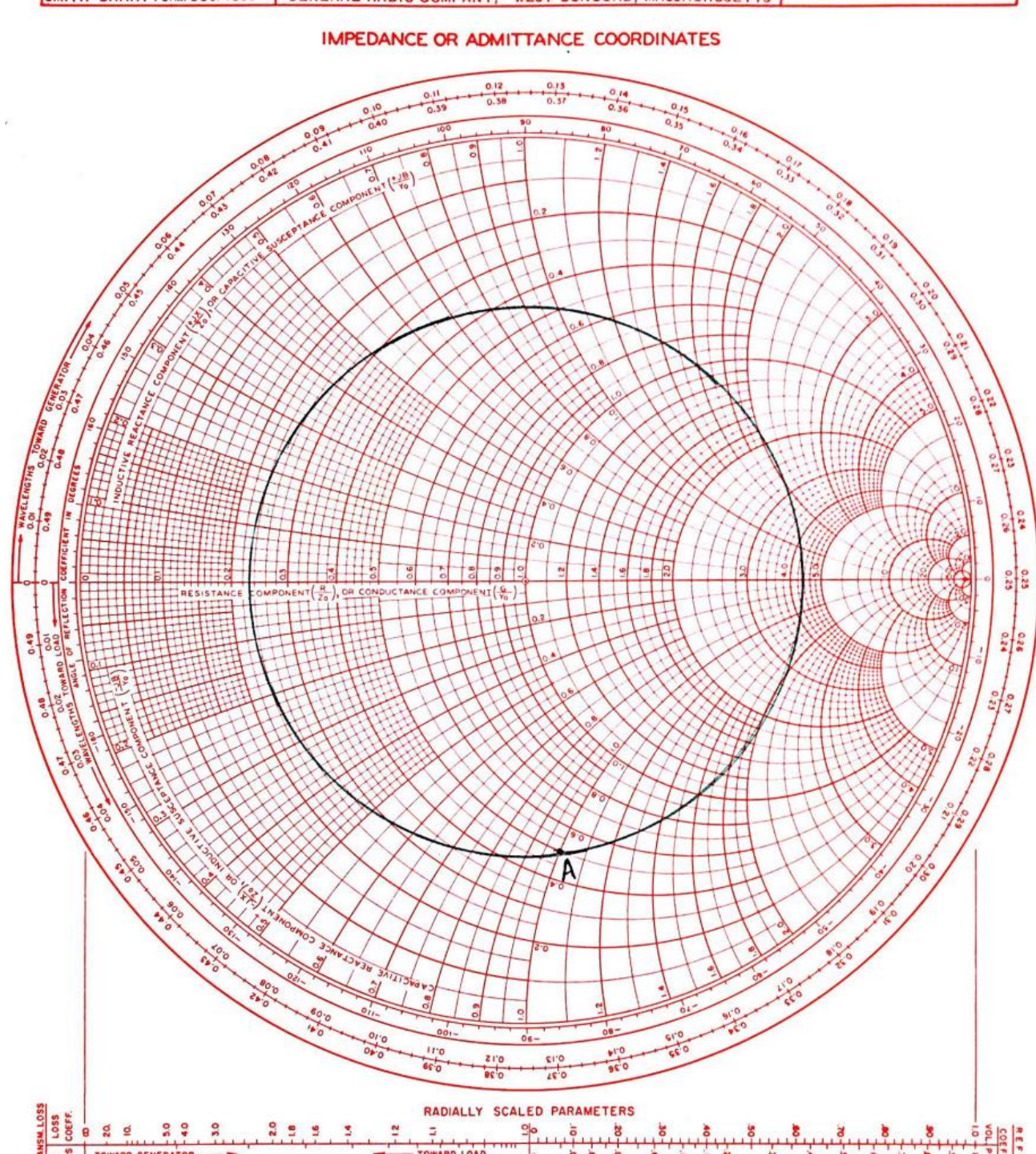
IMPEDANCE OR ADMITTANCE COORDINATES



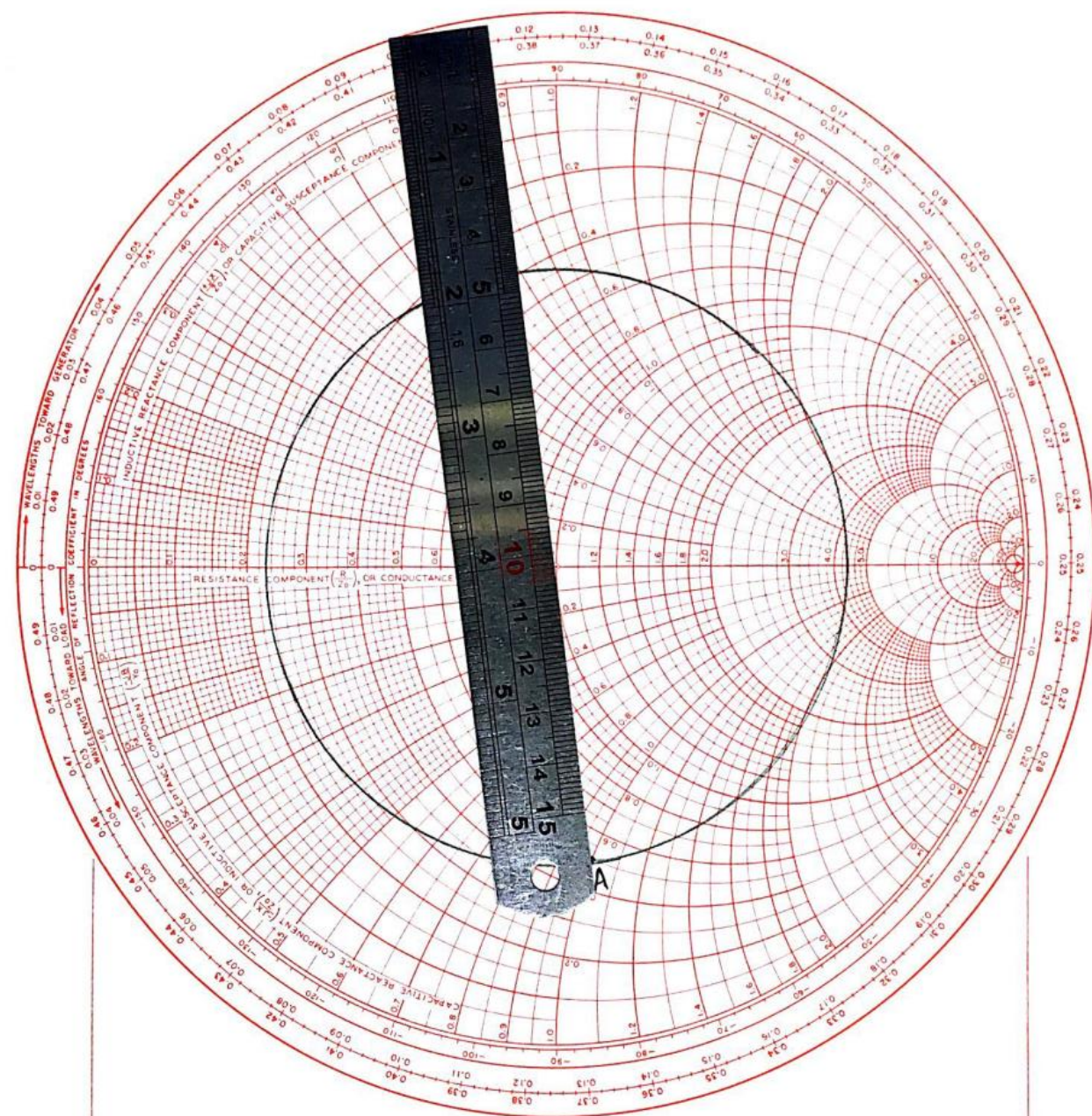
The normalized load impedance is

$$z_L = \frac{Z_L}{Z_0} = \frac{25 - j50}{50} = 0.5 - j1$$

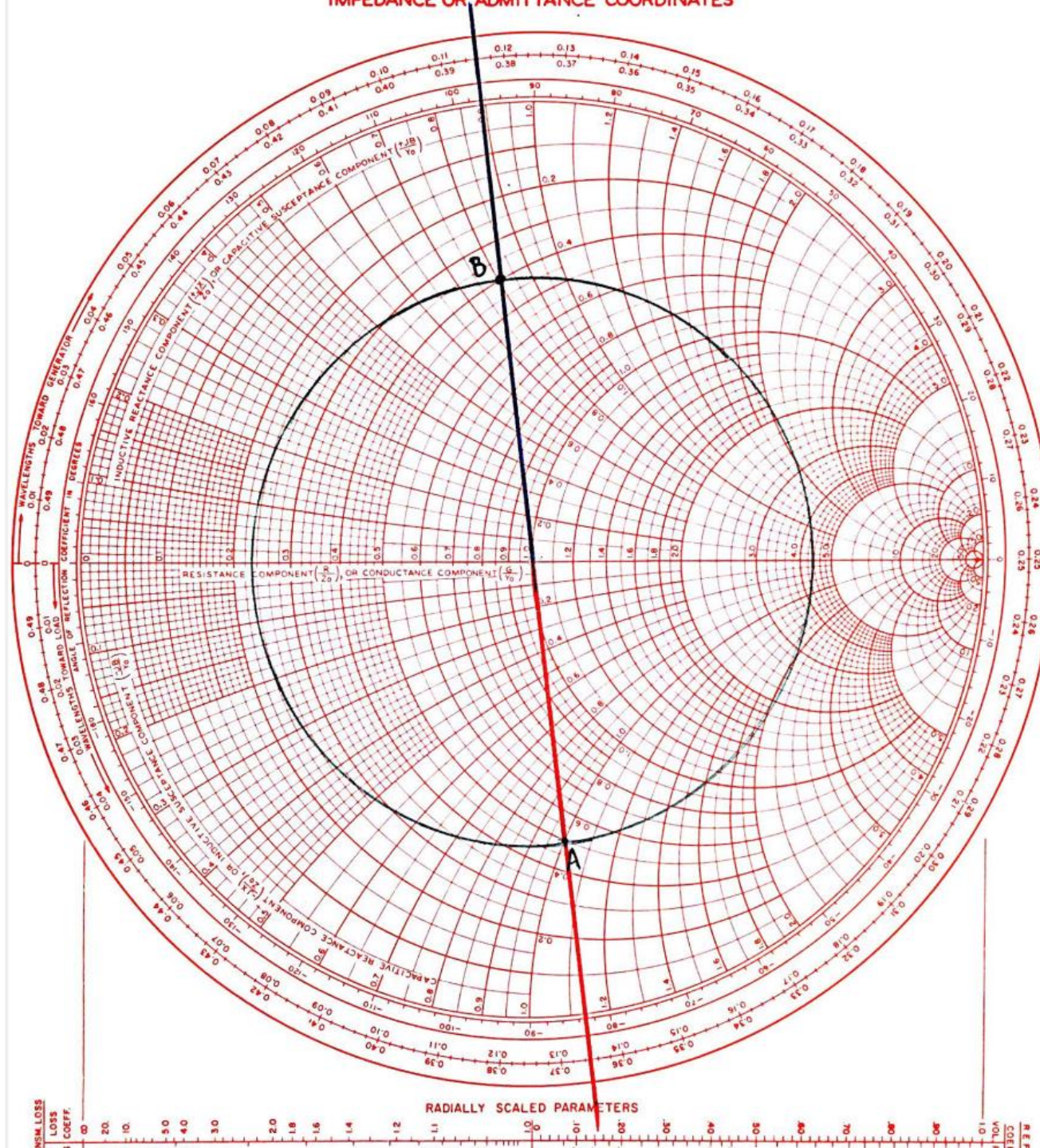
SWR circle



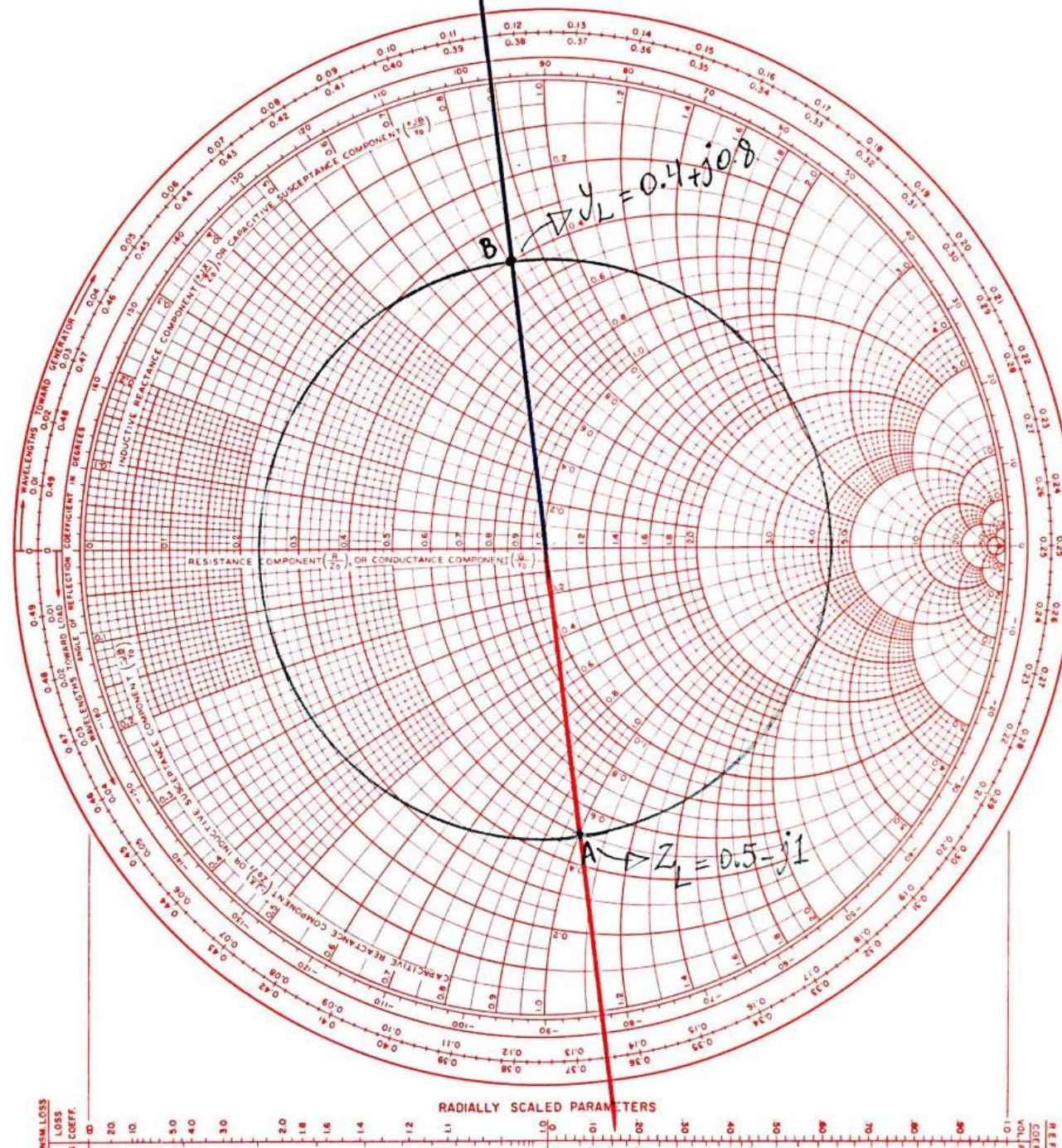
IMPEDANCE OR ADMITTANCE COORDINATES



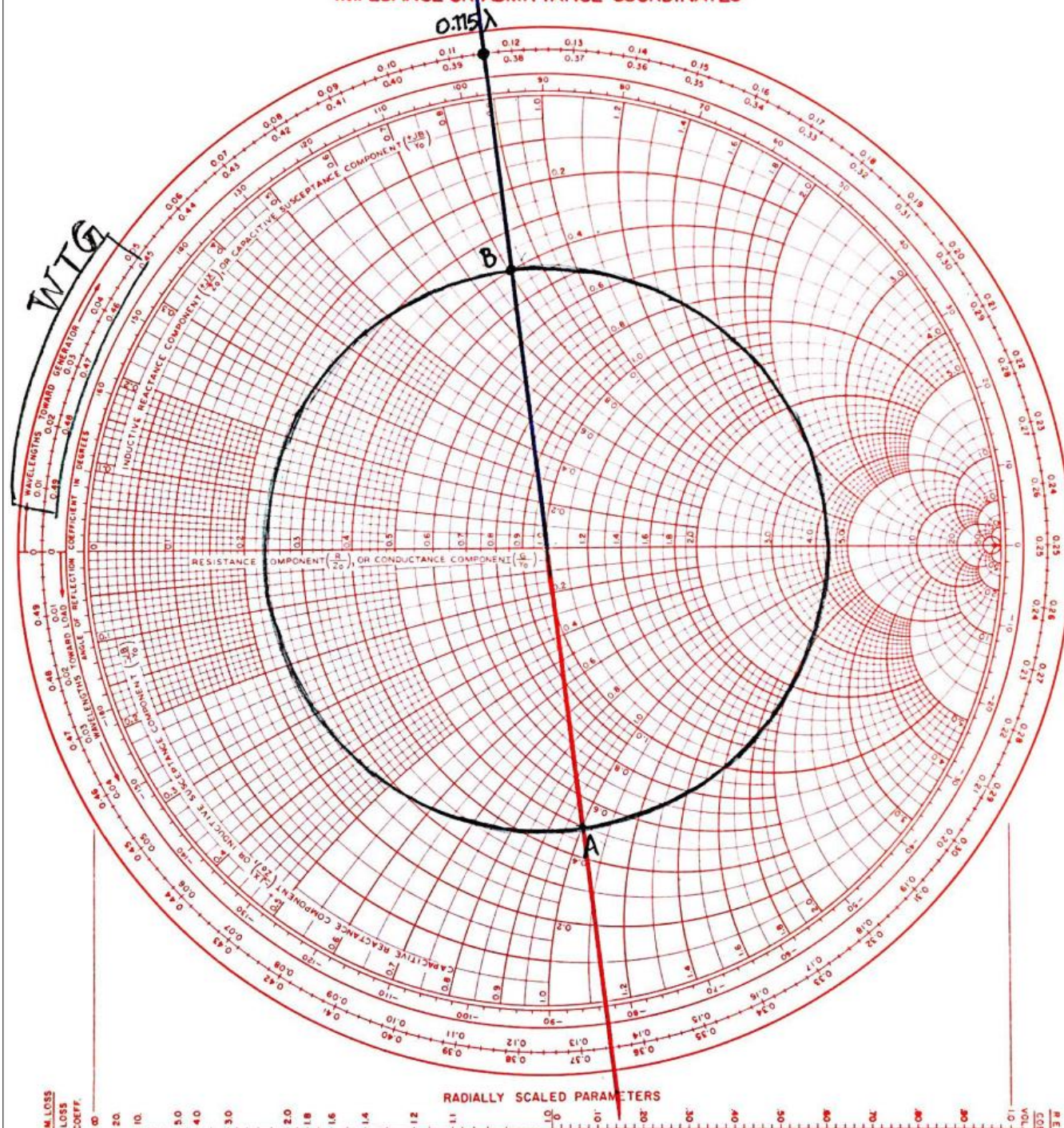
$$y_L = 0.4 + j0.8$$



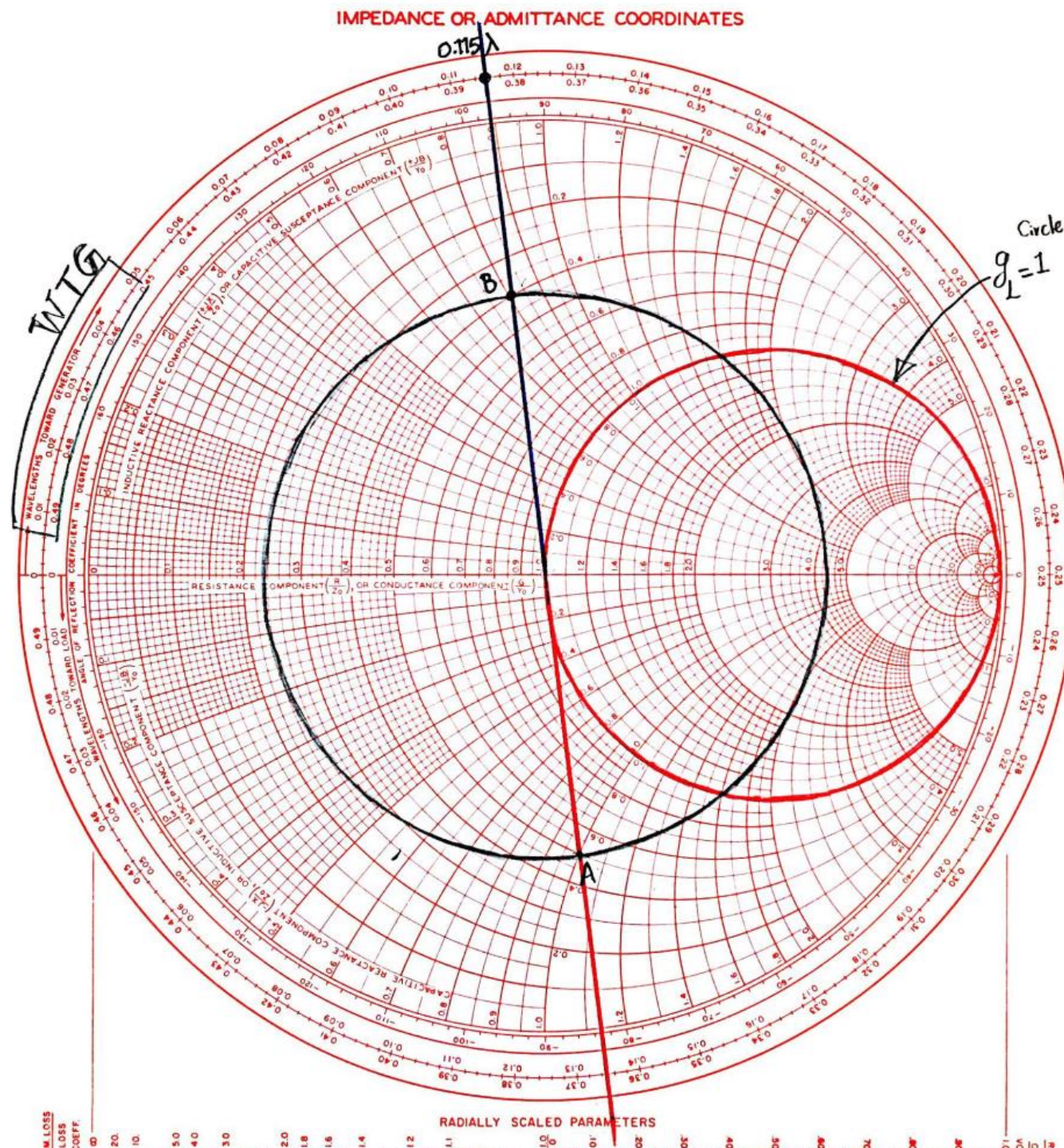
IMPEDANCE OR ADMITTANCE COORDINATES



IMPEDANCE OR ADMITTANCE COORDINATES

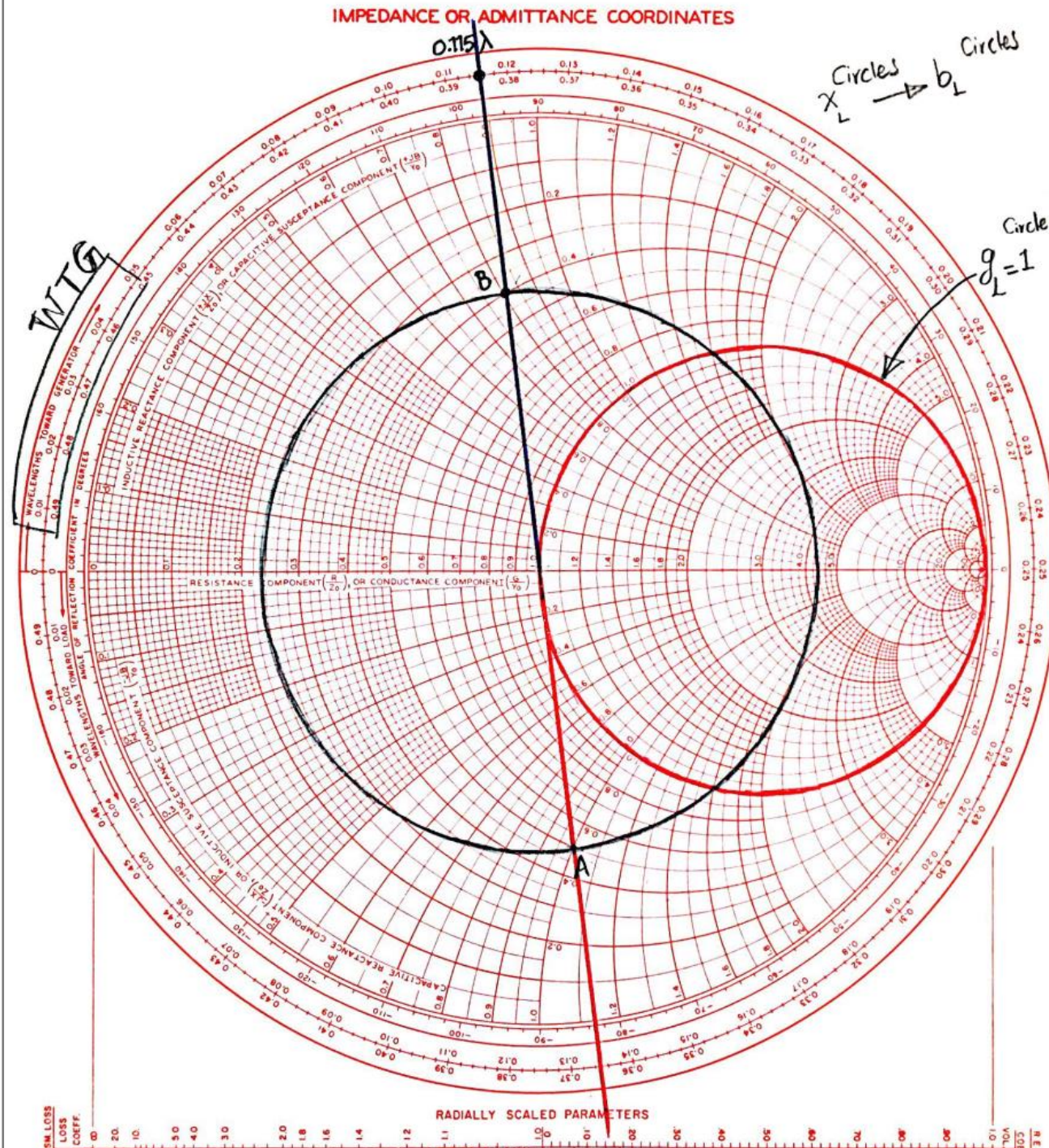


We use
admittance
form

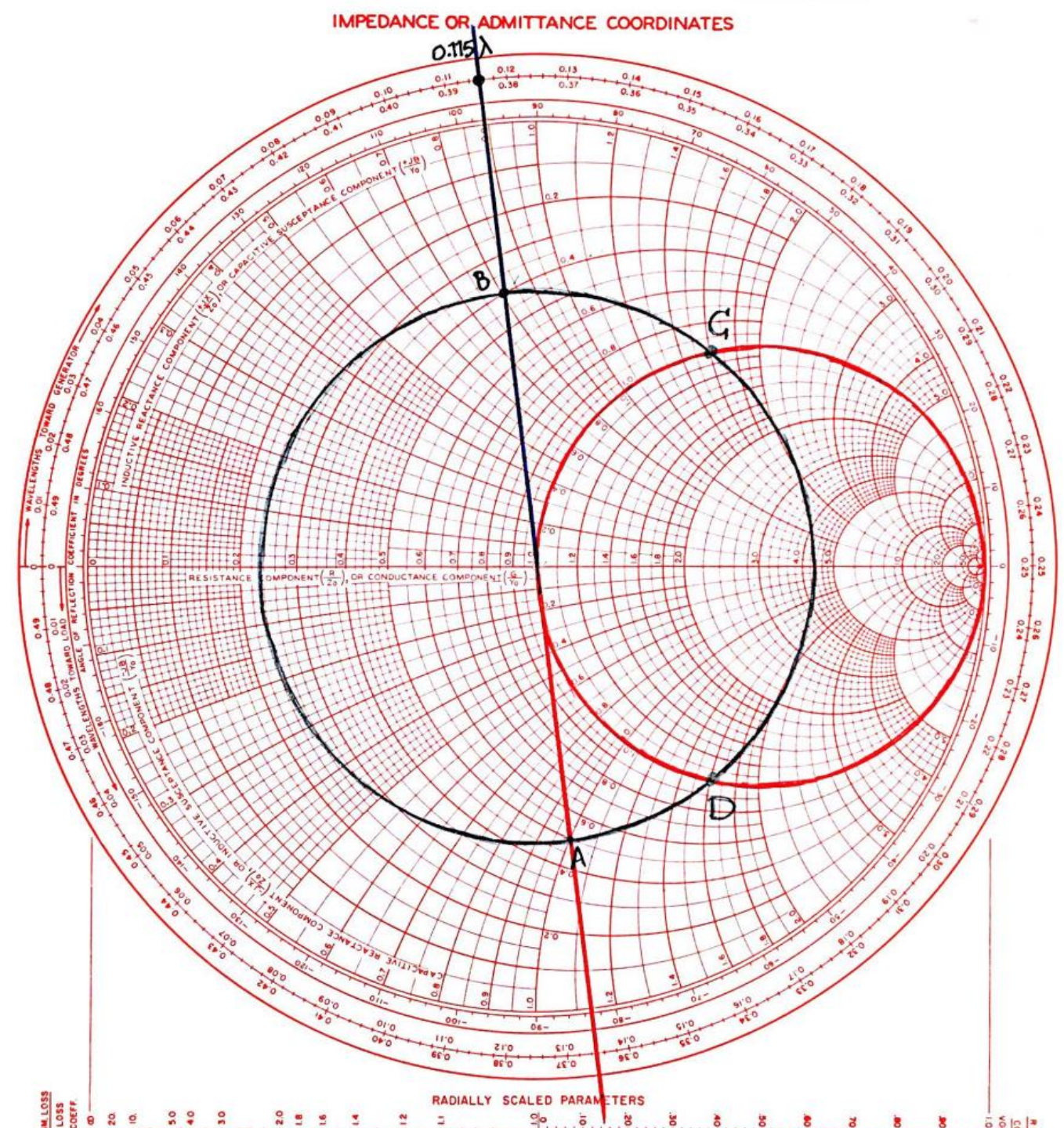


The r_L
circles
becomes
 g_L circles.

To achieve matching, we need to move from the load toward the generator a distance d such that the normalized input admittance y_d of the line terminated in the load has a real part of 1.

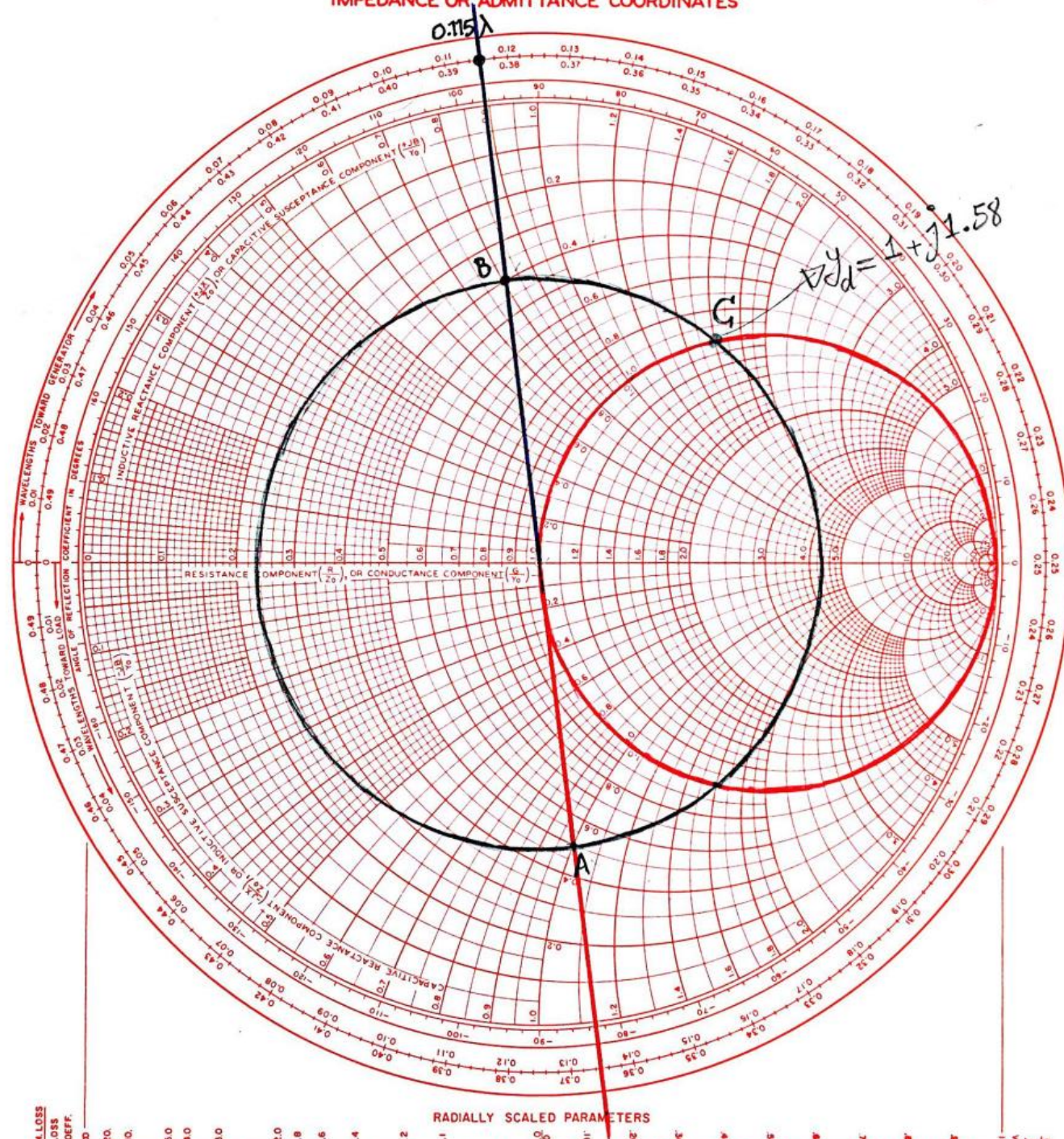


This condition is satisfied by either of two *matching points* C and D.

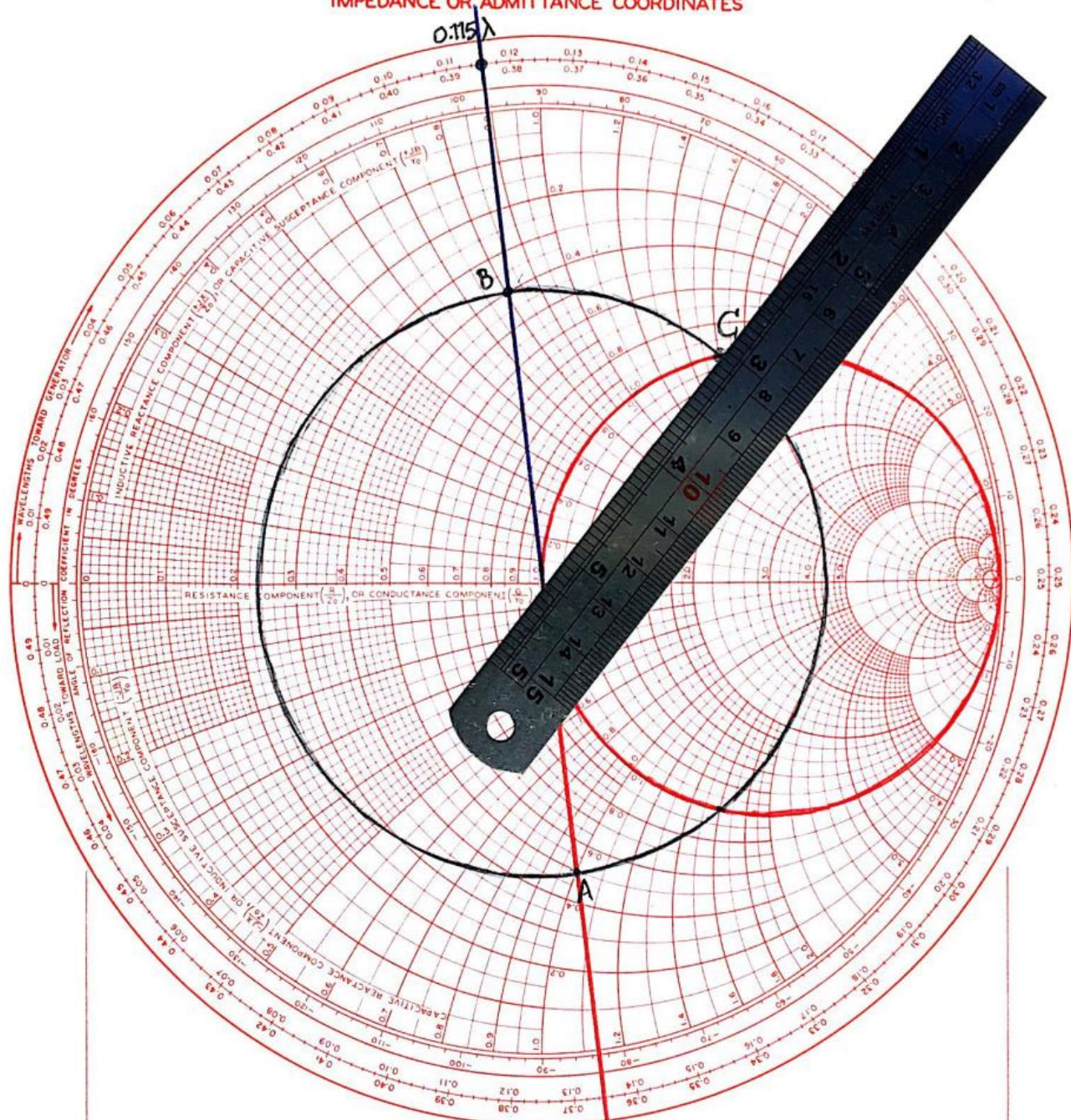


Solution for Point C

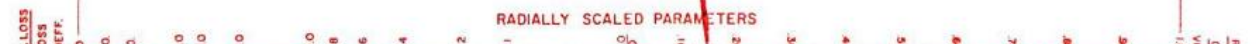
IMPEDANCE OR ADMITTANCE COORDINATES



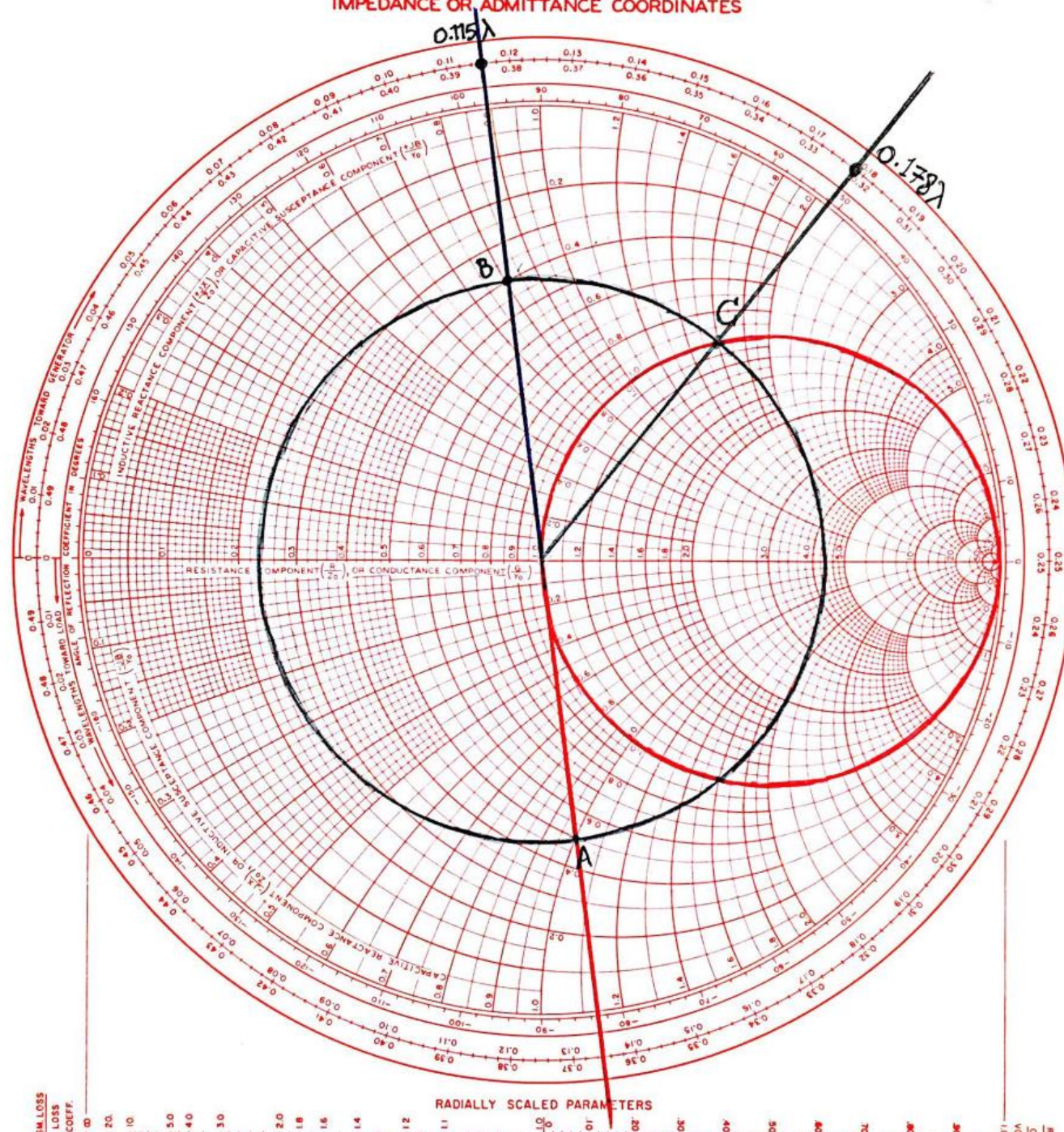
IMPEDANCE OR ADMITTANCE COORDINATES

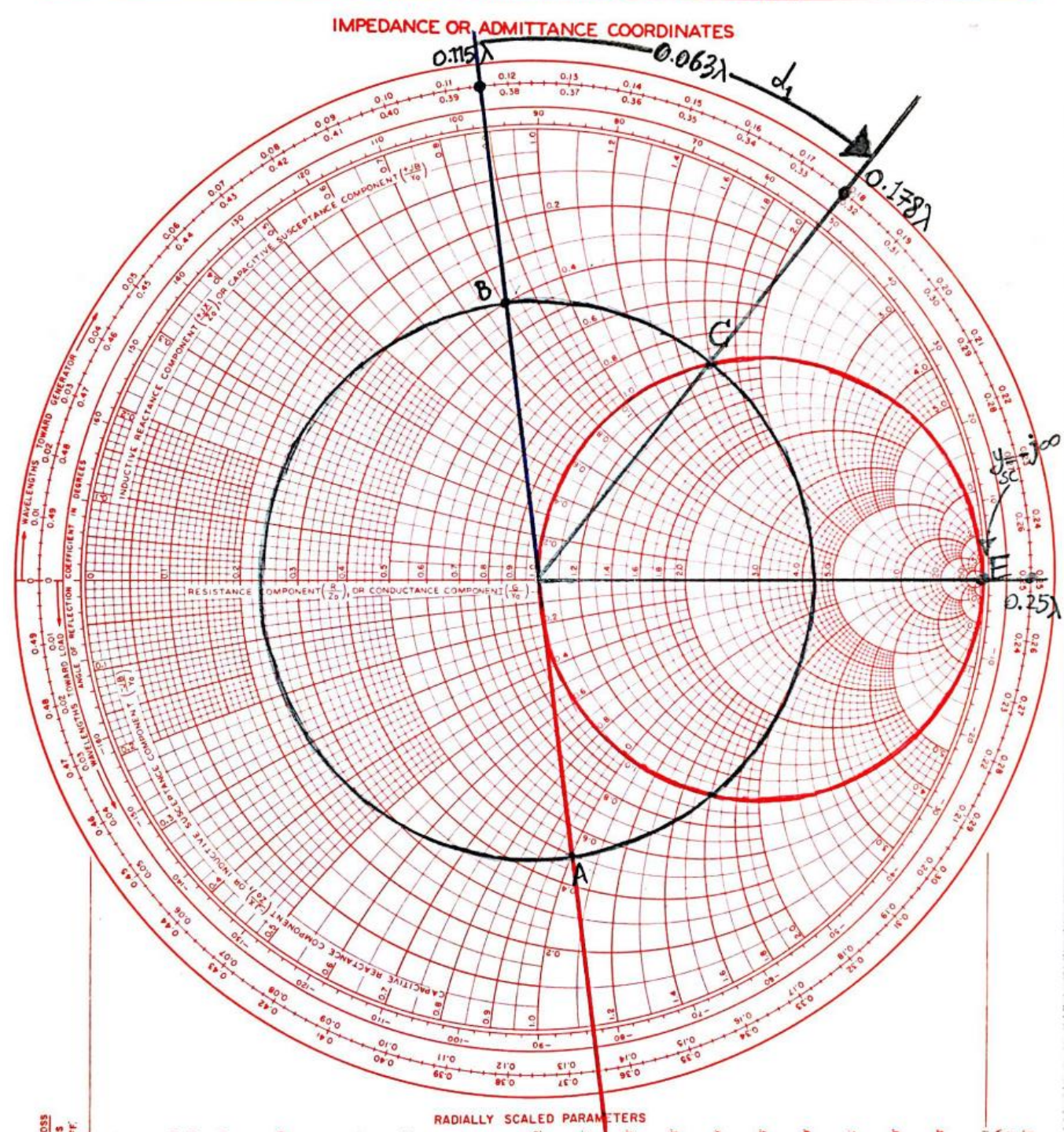


RADIALLY SCALED PARAMETERS



IMPEDANCE OR ADMITTANCE COORDINATES

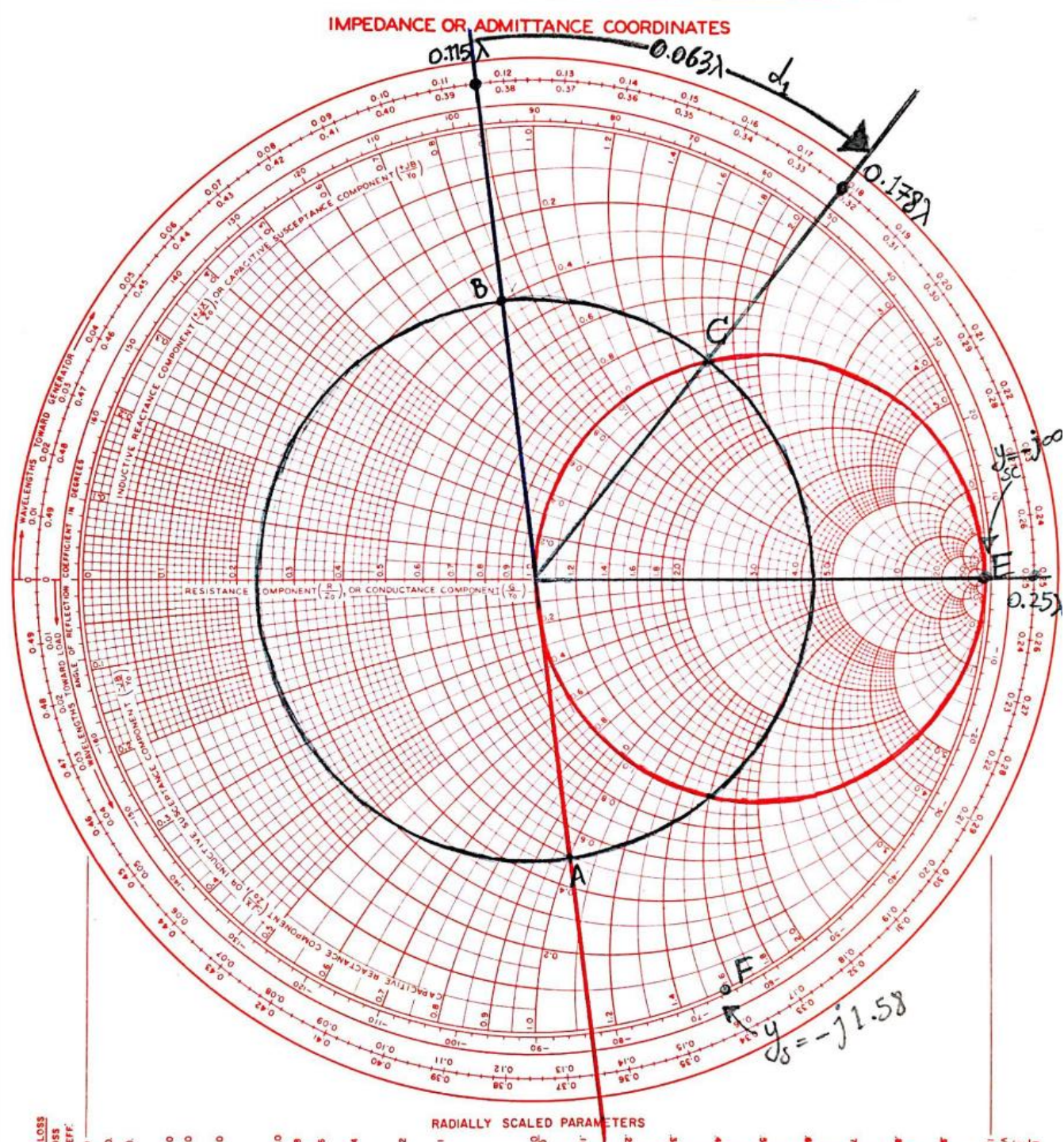




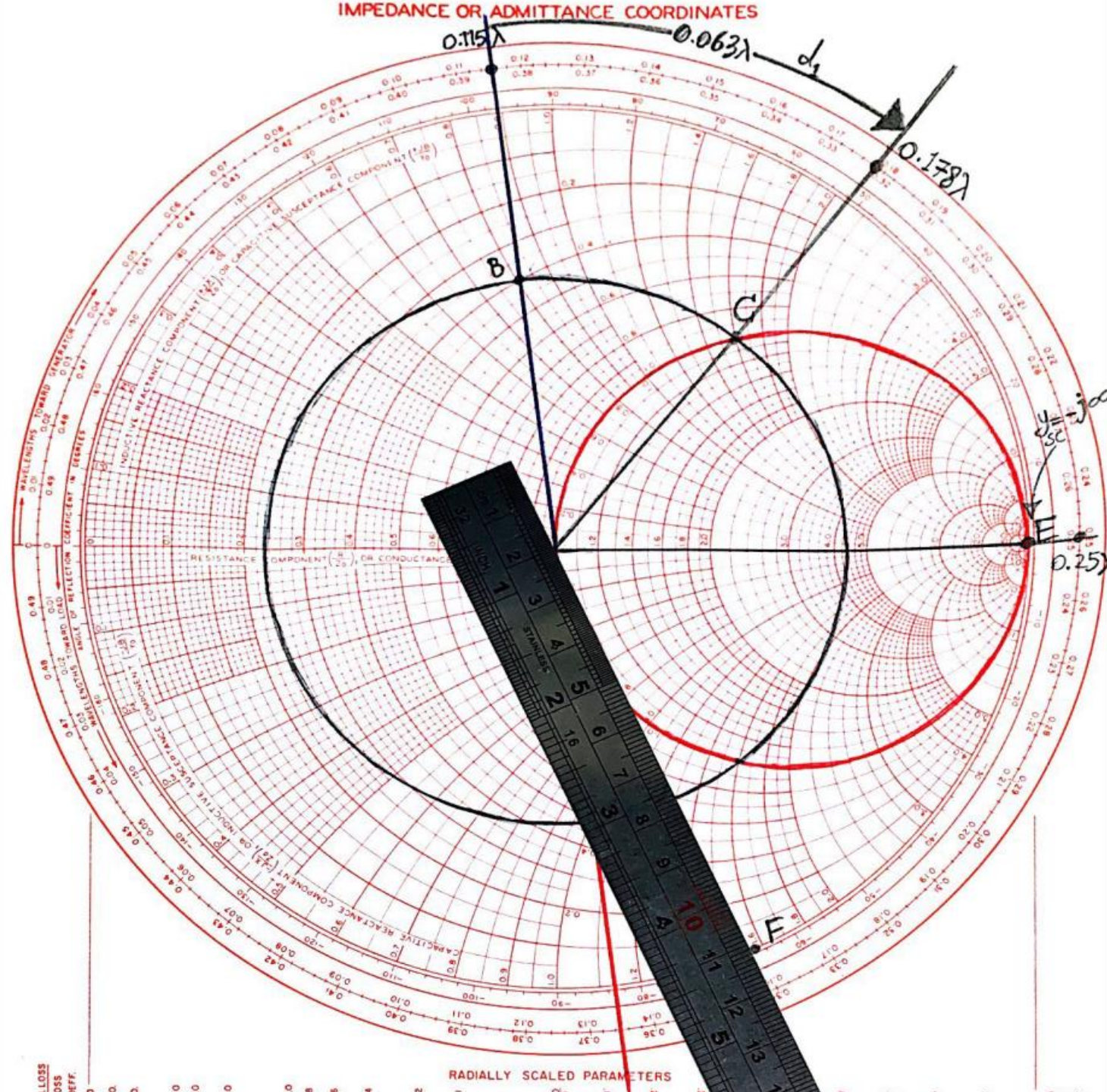
$$y_{\text{in}} = y_s + y_d,$$

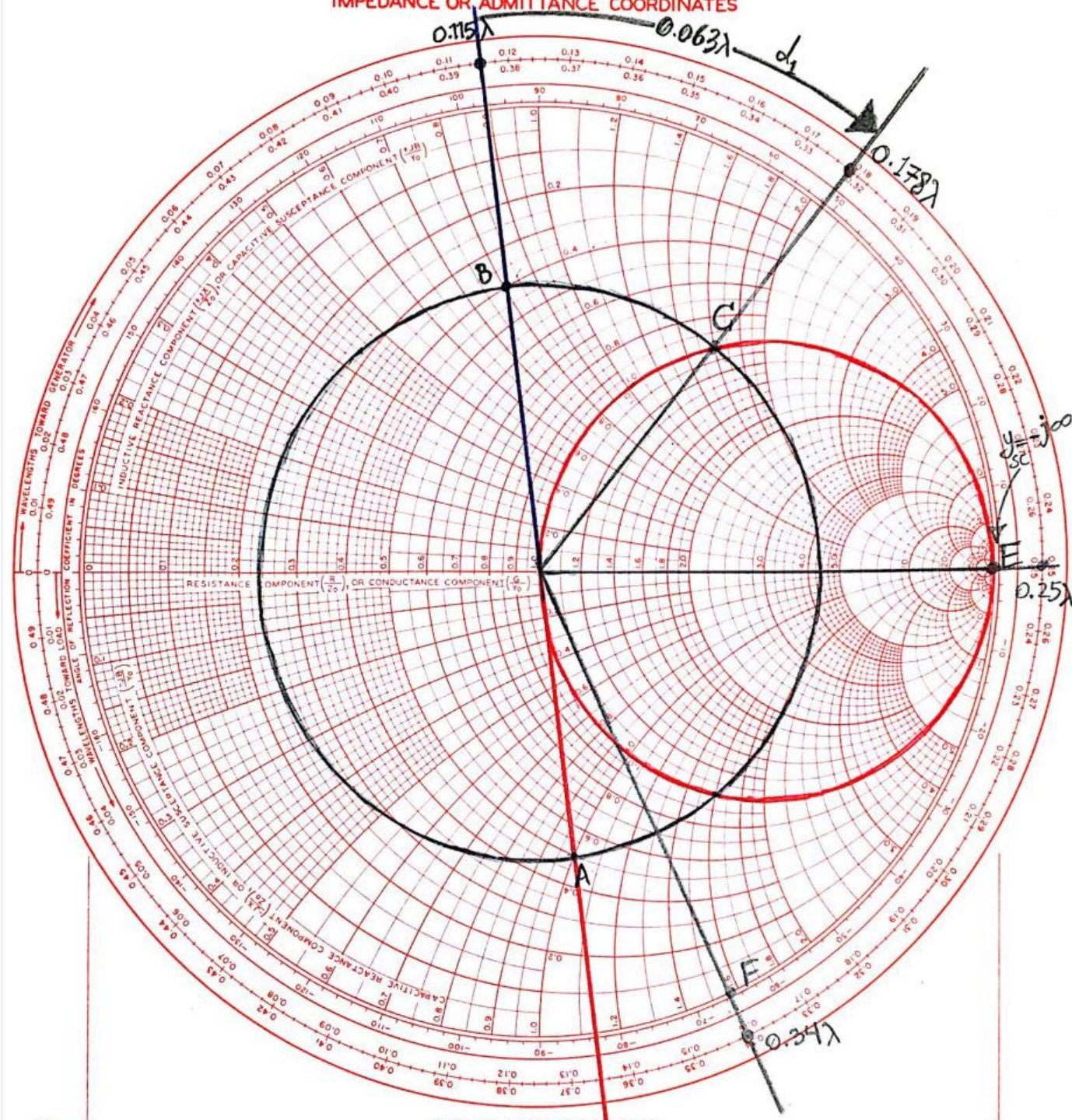
$$1 + j0 = y_s + 1 + j1.58$$

$$y_s = -j1.58.$$

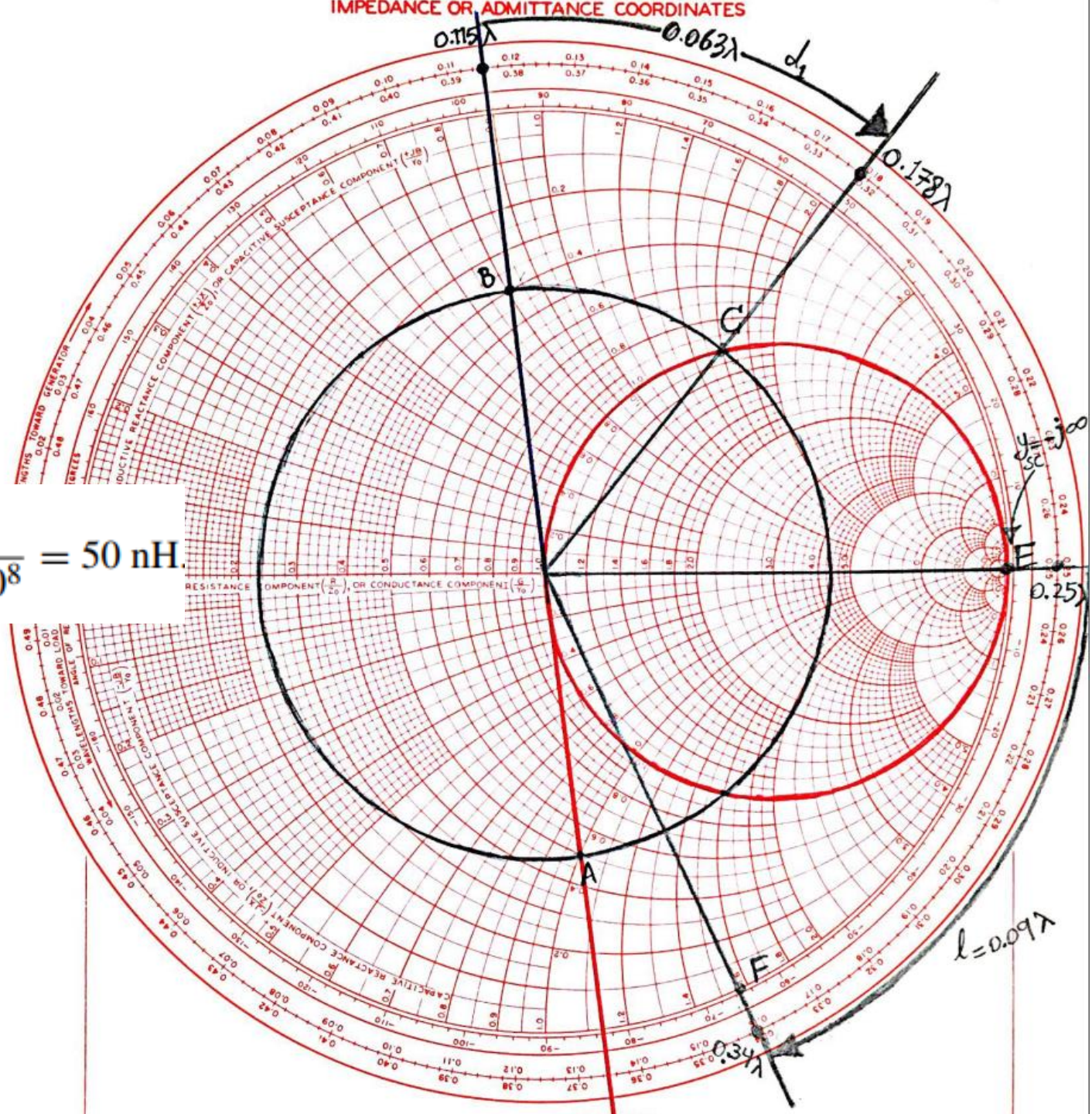


IMPEDANCE OR ADMITTANCE COORDINATES

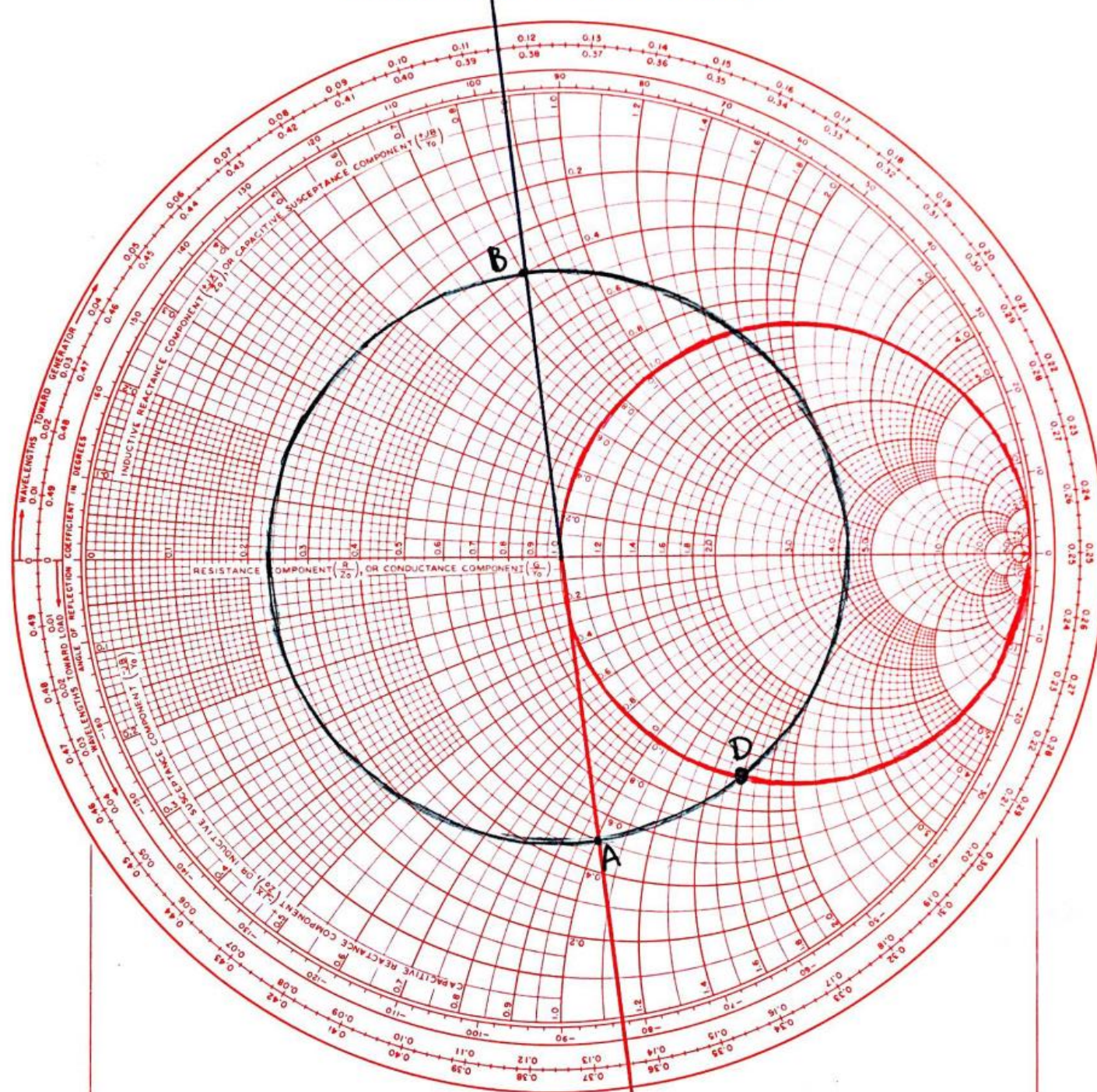




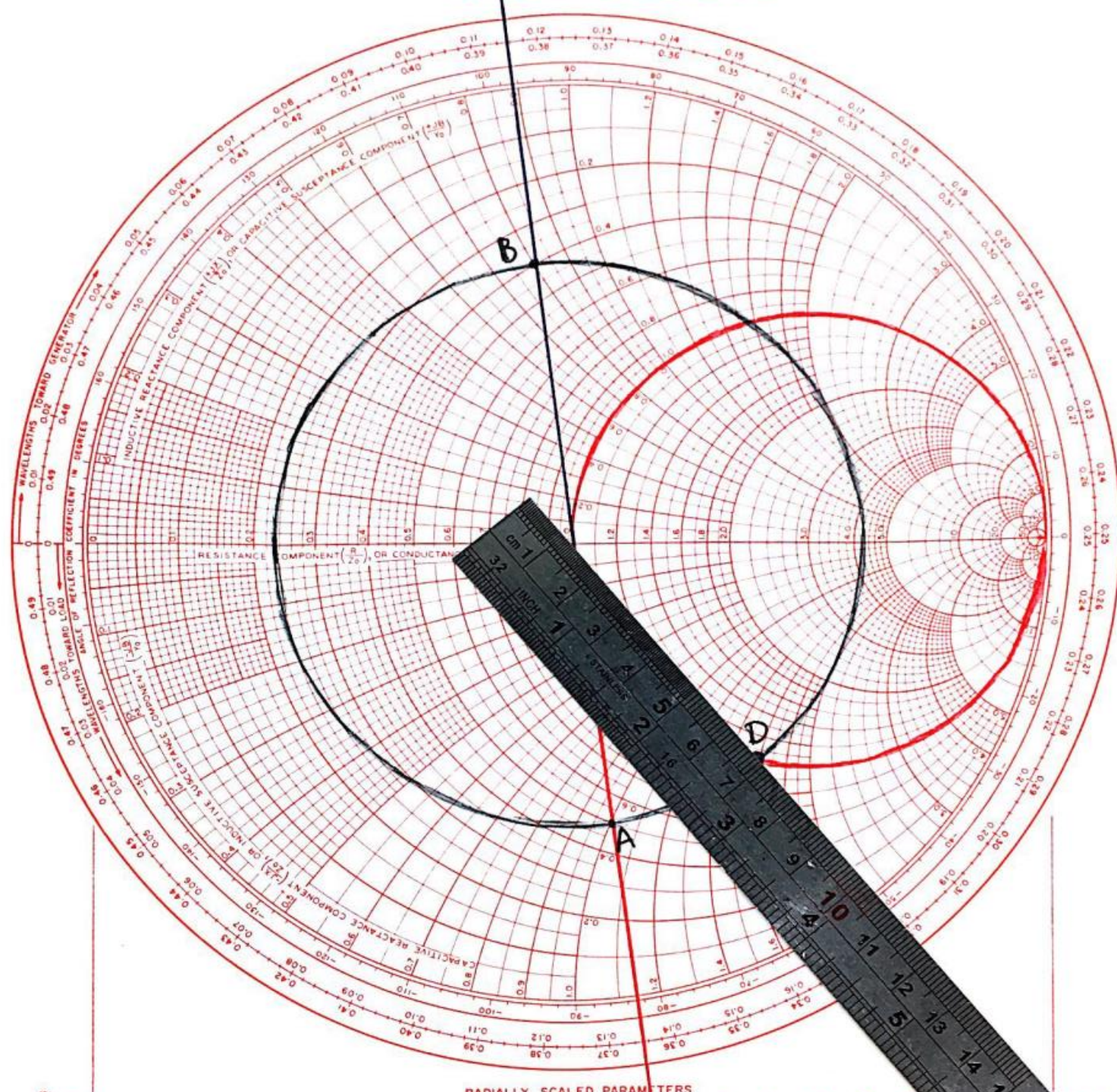
$$L = \frac{31.62}{\omega} = \frac{31.62}{2\pi \times 10^8} = 50 \text{ nH.}$$



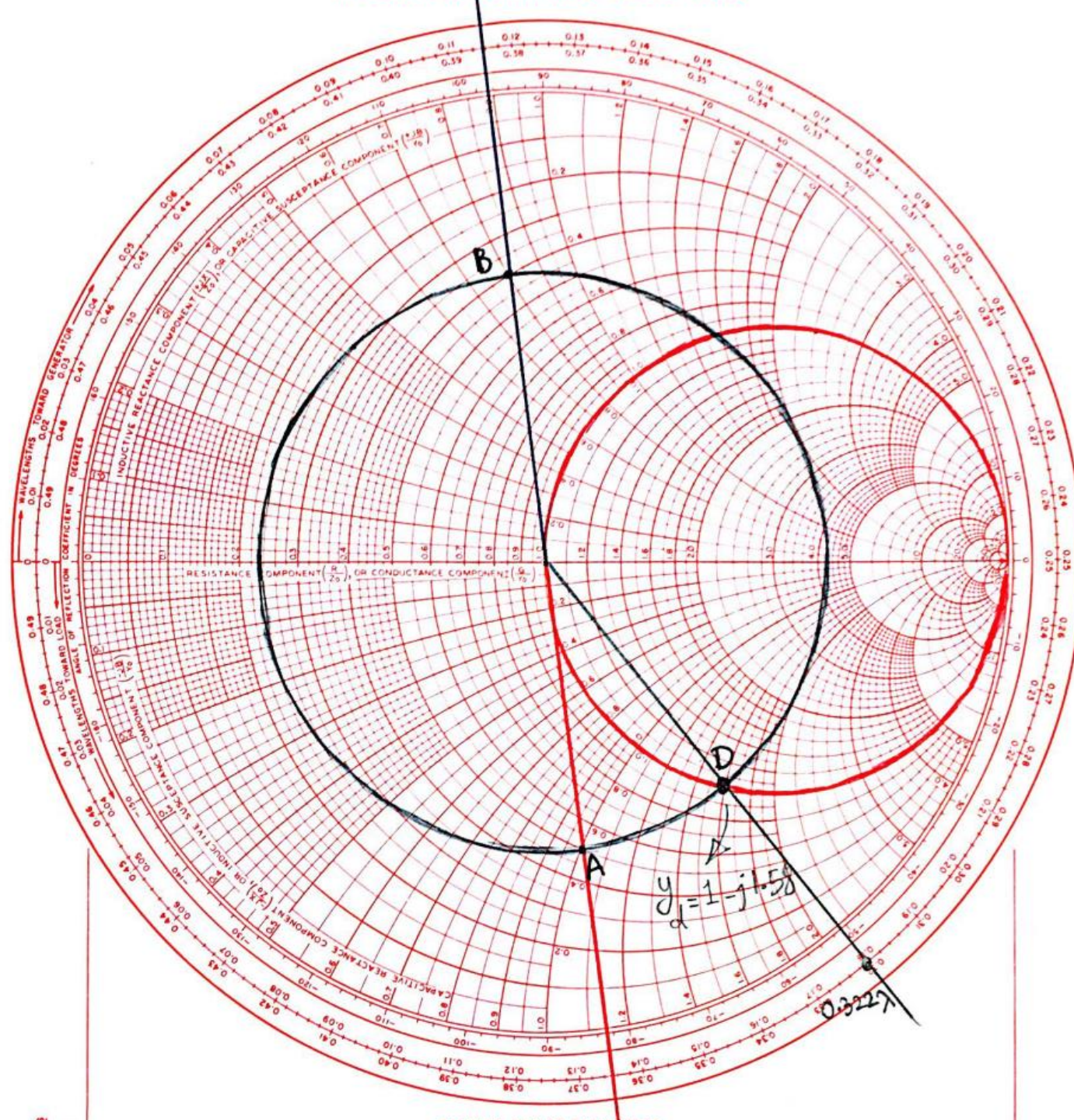
Solution for Point D

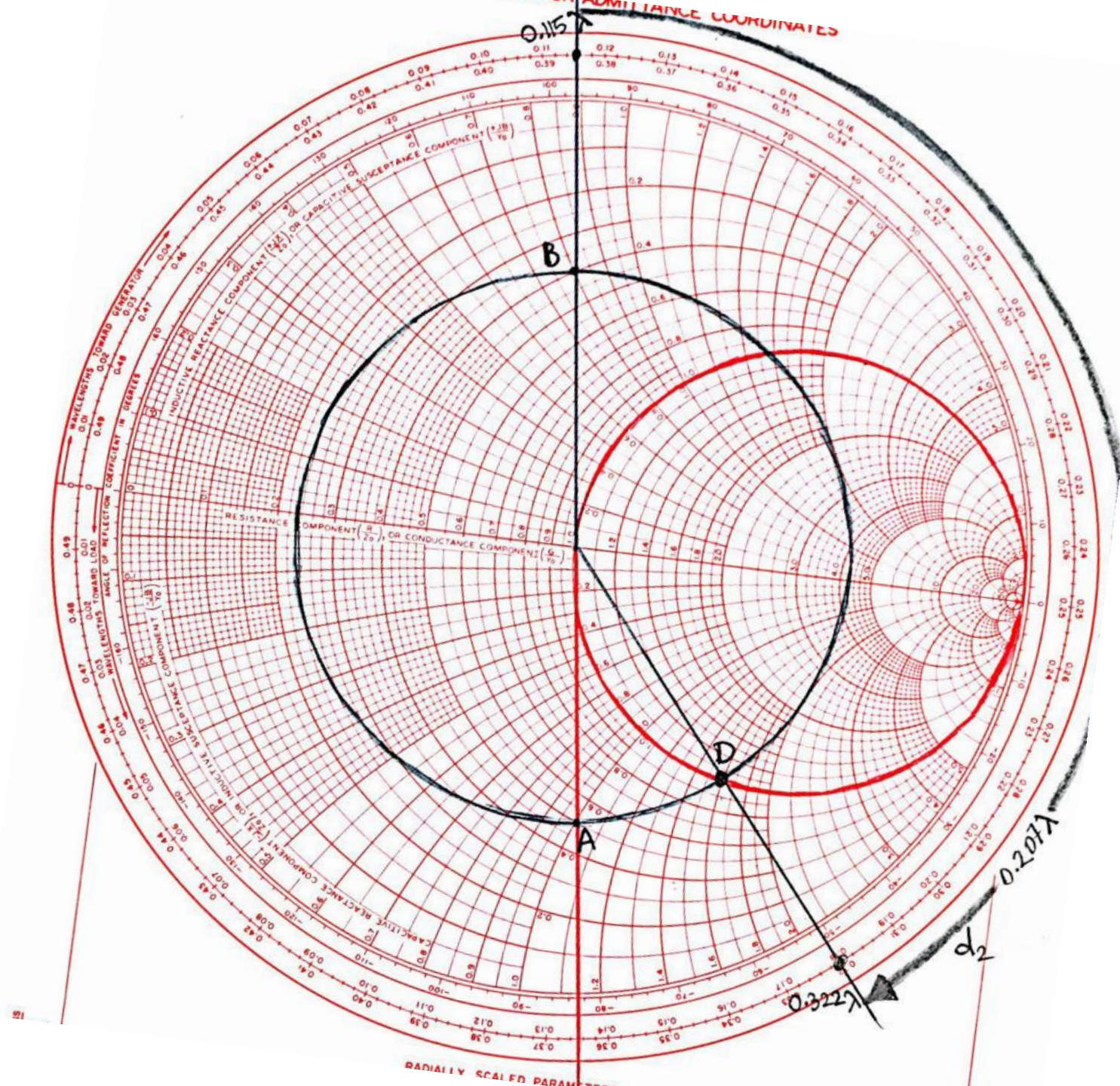


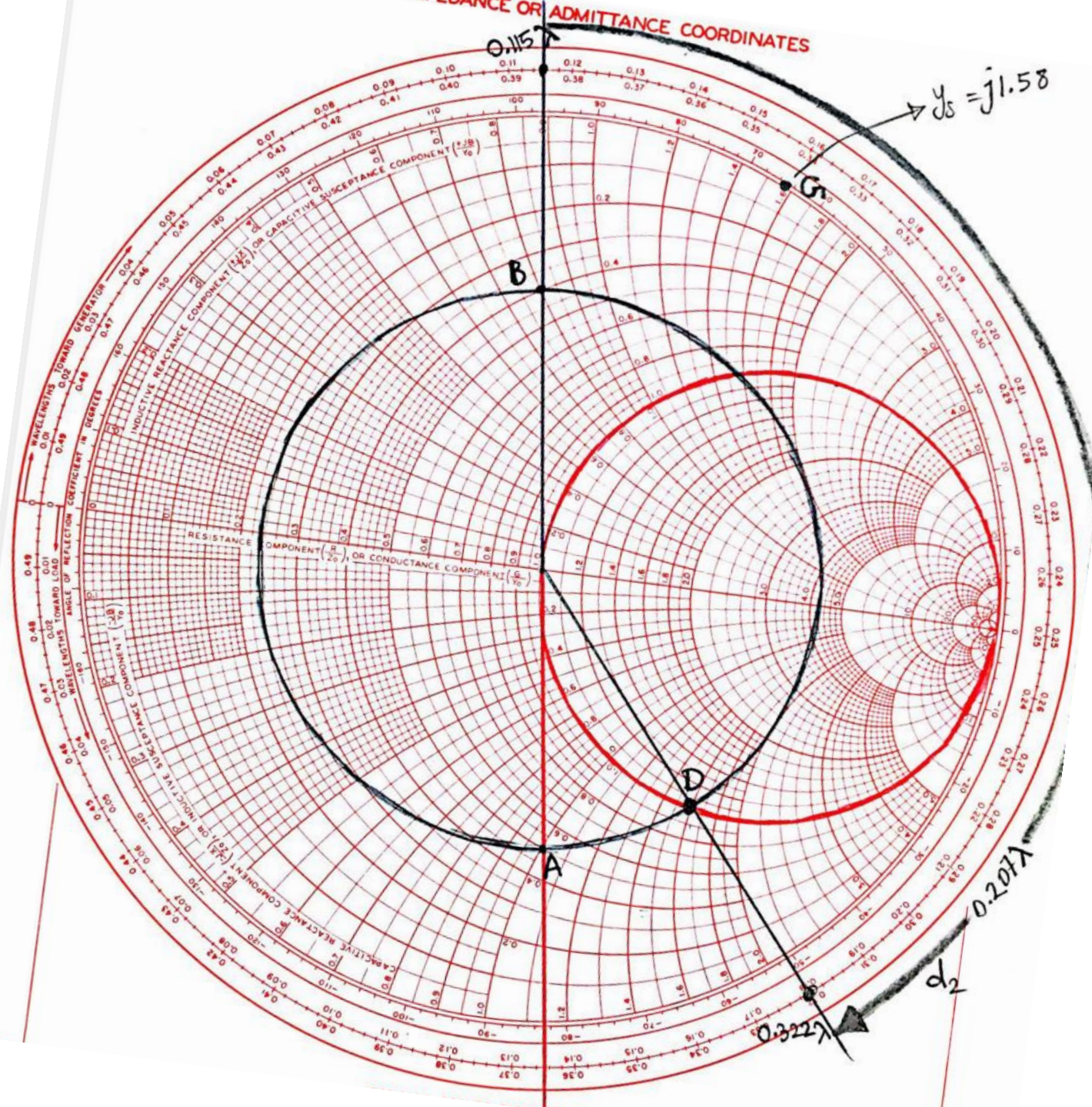
RADIALLY SCALED PARAMETERS

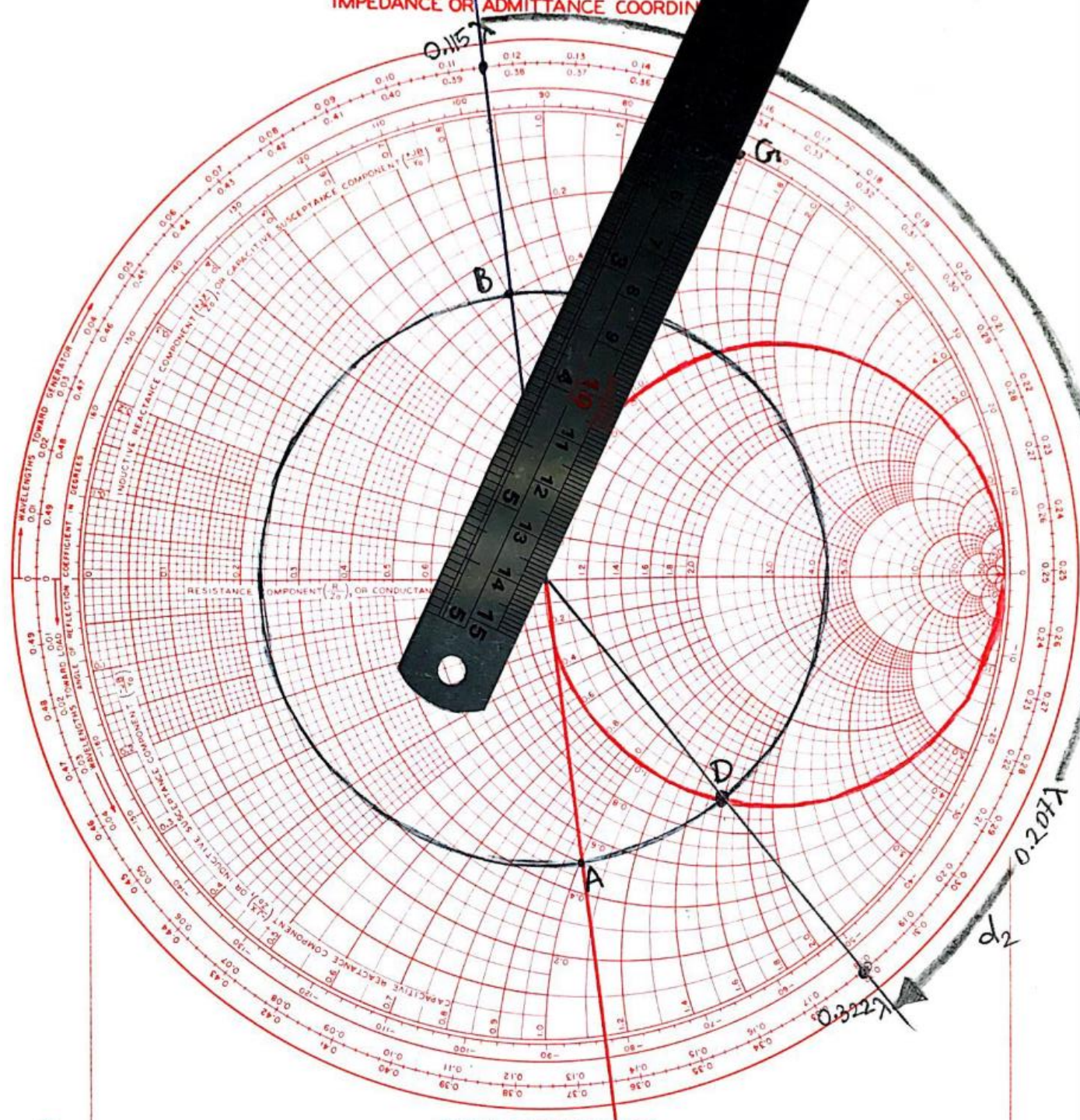


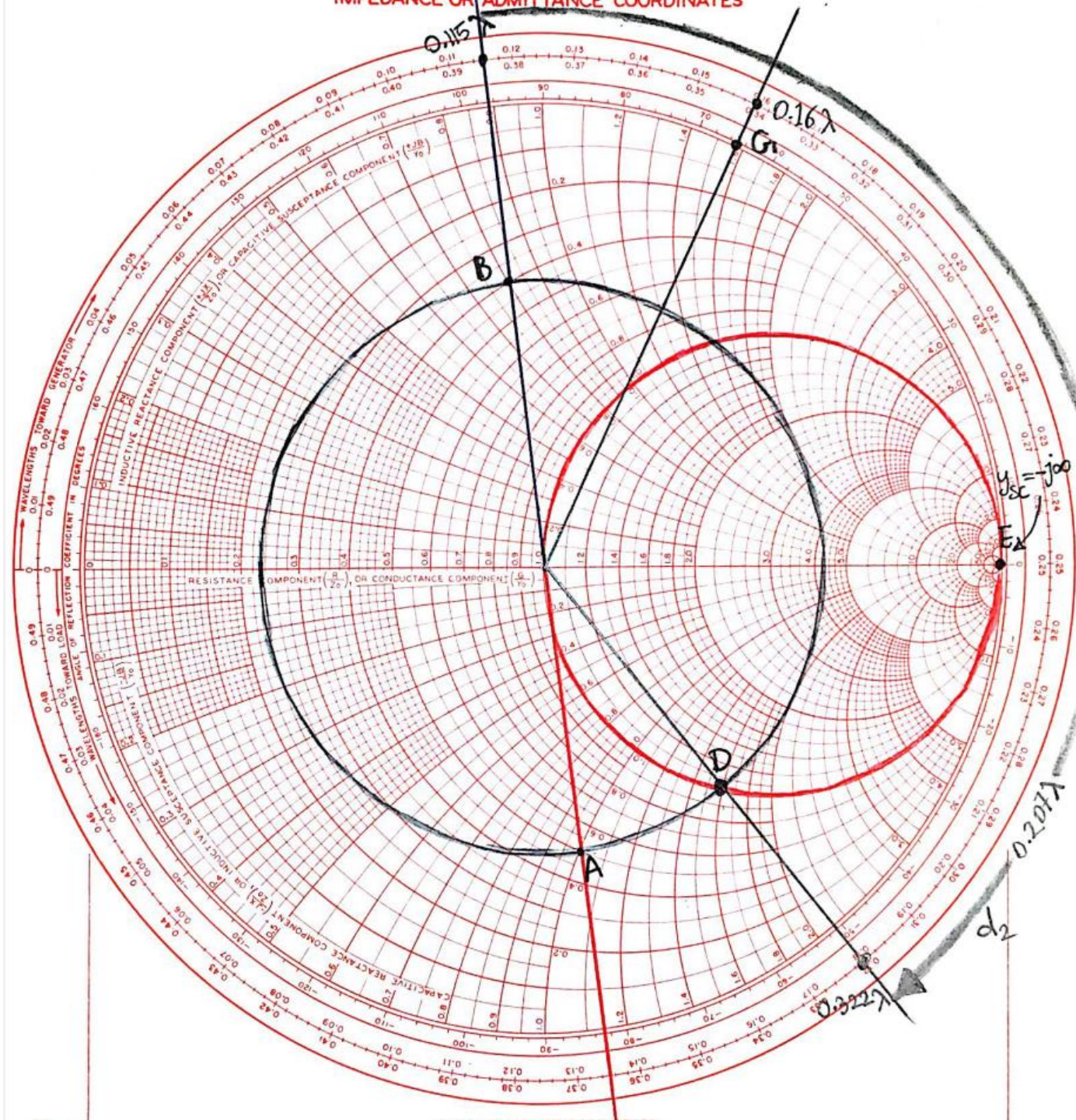
RADIALLY SCALED PARAMETERS





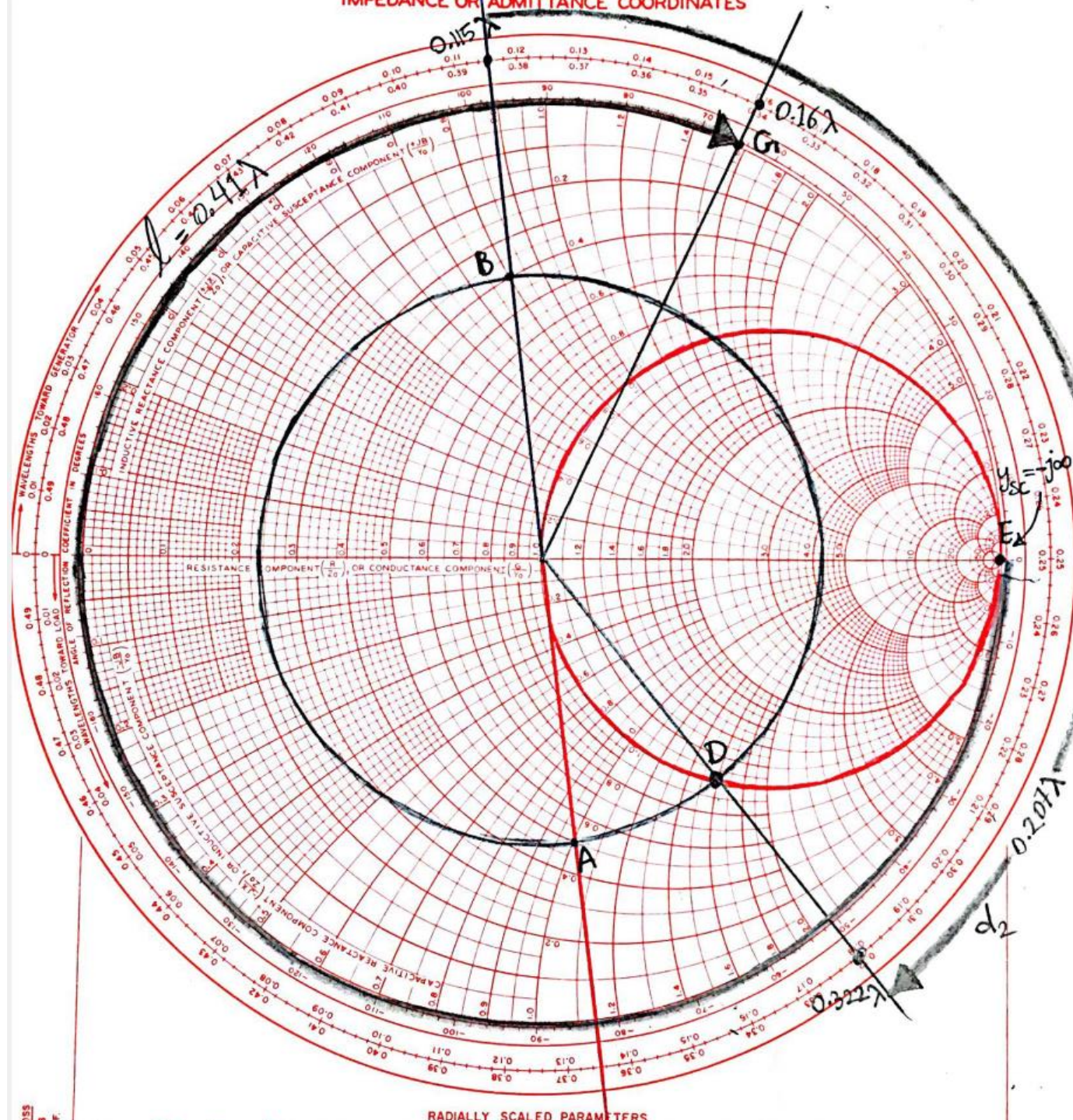






$$Z_{s_2} = -j31.62 \, \Omega$$

$$C = \frac{1}{31.62\omega} = 50 \text{ pF}$$



A load impedance $Z_L = 25 - j50 \, \Omega$ is connected to a $50 \, \Omega$ transmission line. Insert a shunt element to eliminate reflections towards the sending end of the line. Specify the insert location d (in wavelengths), the type of element, and its value, given that $f = 100 \text{ MHz}$.

	d (in λ)	y_d (normalized)	Y_d	y_s (normalized)	Y_s	l (in λ)	y_{in} (normalized)	Y_{in}
Solution #1								
Solution #2								